Fast response MPPT charger for the Técnico Solar Boat

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Introduction to the Research in

Electrical and Computer Engineering

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I declare that this document is an original work of my own authorship and that it fulfils all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Acknowledgments

Agradecimentos... por escrever

Abstract

Solar-powered vehicles rely on solar energy to charge their batteries and maximize their range, particularly solar-powered boats need an effective energy conversion in rapidly changing environmental conditions. In this context, the Maximum Power Point Tracker (MPPT) charger plays a critical role in energy conversion, by ensuring that the solar panels operate on their Maximum Power Point (MPP) while charging a battery. This work focus on the study, design, simulation and test of a fast response MPPT charger for Técnico Solar Boat (TSB).

This document contains the State-of-the-Art review covering photovoltaic principles, MPPT algorithms, DC-DC converter topologies, charging techniques, processing units, and commercial solutions. Special attention is given to the algorithms suitable for dynamic conditions, and DC-DC topologies.

Based on the conduct study and the project requirements, a solution was proposed capable of effectively convert energy, operating the solar panels on their MPP, and without compromising the product cost. Preliminary simulations were conduct to validate the software tools chosen and gain familiarity with the given tools. The ultimate objective of this project is to develop a reliable, efficient, and customizable MPPT solution that improves energy extraction, enhances system monitoring through CAN communication, and contributes to the overall performance and competitiveness of the TSB.

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Keywords: MPPT, MPPT topology, DC-DC Converter, Solar energy, P&O, Solar Boat

Contents

List of Figures

List of Tables

Acronyms

ASIC	Application-Specific Integrated Circuit
CAN	Controller Area Network
CC	Constant Current
CCM	Continuos Conduction Mode
CV	Constant Voltage
DC	Direct Current
DCM	Discontinuous Conduction Mode
ESL	Equivalent Series Inductance
ESR	Equivalent Series Resistance
FPGA	Field Programmable Gate Array
GUI	Graphical User Interface
IC	Integrated Circuit
IncCond	Incremental Conductance
MCU	Microcontroller Unit
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracker
LUT	Look-Up Table
OC	Over Current
OTS	Off the shelf
P&O	Perturb and Observe
PCB	Printed Circuit Board

PV Photovoltaic

R&D Research and Development

SG01 São Gabriel 01

SR03 São Rafael 03

STC Standard Testing Conditions

TSB Técnico Solar Boat

1 Introduction

1.1 Motivation

With environmental pollution intensifying worldwide, the transition to cleaner energy sources such as solar power has become increasingly urgent. In 1985, only 20.82% of the energy produced in the world came from renewable energies. These numbers have been rising every year, and in 2024, have reached 31.92% [11]. In 2024, 6.9% of the energy produced in the world comes from solar panels, and in Portugal this number rises to 14.5%, demonstrating the importance and the impact of solar energy [11]. Solar energy still plays a minuscule role, and it is listed behind the other sources of energy in terms of its contribution to meeting the world's energy demand. However, solar energy is becoming more relevant, with the cost of solar panels dropping significantly in the last decade.

In comparison to other forms of green energy, Photovoltaic energy is relevant due to its availability, simplicity, lower maintenance, environmental friendliness, reliability, and many other benefits. More recently, it is becoming more relevant in the automotive industry, with solar-powered cars, boats, and robots [12]. The CO2 emissions of the automotive sector are one of the main contributors to global warming. In 2018, 29% of total CO2 emissions in the EU came from the transport sector, with 4% coming from the maritime sector alone [13]. These challenges have directly motivated the development of the Técnico Solar Boat (TSB) project.

In 2015, TSB was created with the goal of designing and building a solar-powered boat to compete in international competitions. Since then, the project has grown, and several vessels have been built. It began with the construction of the first solar prototype, São Rafael 01, which still had a lot of room for improvement and was followed by São Rafael 02 and 03. The team then decided to approach the hydrogen energy and autonomous driving, and therefore São Miguel 01 and 02 and São Pedro 01 were built. More recently, a vessel was built to combine all the technologies used before and was named São Gabriel 01 (SG01).

All these prototypes used solar energy to maximize their range and efficiency. In the first years the energy produced was small, and the whole system was built using commercial Off the shelf (OTS) components. In 2020, the team started to build its own solar panels for São Rafael 03. Many other systems were also designed and built in house, but there is still one system that is yet to be developed, the Maximum Power Point Tracker (MPPT) charger.

The MPPT charger is a fundamental subsystem in photovoltaic energy systems, responsible for maximizing the power extracted from solar panels by ensuring continuous operation at the Maximum Power Point (MPP). A MPPT charger is typically a power converter that continuously monitors the voltage and current produced by the photovoltaic modules and adjusts their electrical operating point accordingly. By dynamically regulating these operating parameters, the MPPT compensates for variations in environmental conditions such as solar irradiance and temperature, which cause the MPP to shift over time. As a result, the photovoltaic system can operate with improved efficiency and deliver maximum available power under changing conditions [14].

1.2 Objectives

This project aims to design and implement a MPPT charger for the solar panels used in TSB project. The MPPT charger will convert the energy produced by the solar panel as efficiently as possible with the use of a quality DC-DC converter and the implementation of MPP tracking algorithms.

The main objectives of this project are:

- Study and understand the operation of solar panels and MPPT techniques;
- Choose the most suitable DC-DC topology for the system;
- Implement a fast response and accurate control algorithm;
- Design and simulate the hardware and software;
- Implement the system in hardware and develop a Printed Circuit Board (PCB);
- Test and validate the performance of the circuit;
- Ensure the safety and reliability of the MPPT charger for its integration in the TSB project;
- Provide performance data of the solar panel and the MPPT through a Controller Area Network (CAN) communication interface.

By achieving these objectives, the project will contribute to a more efficient solar system, maximizing the energy received from the solar panels and increasing the available data for performance tracking of the MPPT and solar panel. This data will be used to later improve the energy efficiency of the MPPT and take conclusions about the manufacture quality of the solar panels build by TSB project. A cheap, efficient and versatile MPPT will also mean future savings to the team, since it could be used in a variety of vessels.

1.3 Outline

This document is organized into five main chapters:

- **Chapter 1** introduces the context of the work, presenting the motivation behind the development of an MPPT charger for the TSB project. The objectives of the project are defined, and the scope of the proposed solution is established;
- **Chapter 2** presents the State-of-the-Art related to photovoltaic energy systems and MPPT technology. It reviews the operating principles of solar panels, solar energy production, and the most relevant MPPT algorithms. Additionally, different DC-DC converter topologies, battery charging techniques, processing units, and commercial solutions are analyzed and compared;
- **Chapter 3** describes the proposed solution for the MPPT system. The overall system architecture is presented, followed by a summary of the technical requirements. The adopted design approach and methodology are detailed, including converter selection, control strategy, simulation, and experimental validation plan;
- **Chapter 4** presents the preliminary work developed so far, including initial simulations and early design results that support the feasibility of the proposed solution;
- **Chapter 5** outlines the planning and scheduling of the project, presenting the development timeline and key milestones required to achieve the defined objectives.

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2 System

2.1 Overview

São Gabriel 01 (SG01) is the most recent vessel built by TSB, which was designed to compete in Sea Lab category of Monaco energy boat challenge. To compete in this category a vessel with 8 x 2.5 m was designed creating new challenges for the team. This vessel was designed to use boat hydrogen and solar energy, using hydro foils to hydroplane which highly increases efficiency. Additionally, it will use actuators and sensors to drive autonomously to compete in AI class of Monaco Energy Boat Challenge.

Since it is a larger and heavier boat than the previous ones built by TSB it needs more power than the others. So a motor with 130 KW peak power was chosen. This much power is only possible to provide with high voltages, so for the first time in TSB history this vessel have a High Voltage system (300-600 V) and a Low Voltage system (30-50 V).

The High Voltage system is responsible for the drive train, powering a motor. Since the energy consumed by this motor is much larger than solar panels can ever achieve, the boat is designed to be powered by hydrogen (not installed yet).

As for the low voltage system, it is designed to power all auxiliary system (dashboard, telemetry, flight control, electrical string, throttle, etc.). Some of this auxiliary systems are critical for the normal use of the vessel, for example if the dashboard turns of, the pilot will not know the state of the high voltage system (temperature, voltage, state of charge, etc.). Or even more critical, if the throttle does not work, the boat can not electrically move.

The auxiliary systems are always changing and growing, so it is difficult to estimate the power consumption of this system. For this work it is considered 1000 W of continuous consumption. It is not a perfect estimation because the power consumption will depend on the driving mode (autonomous driving, hydroplaning or normal drive) which can create different power profiles due to the actuators. Additionally, the cooling system (which is planed to be moved to the High Voltage system) have high power consumption too.

To help to maintain the low voltage system always live, we use solar panels which are connected to several Maximum Power Point Trackers (MPPTs).

For better understanding of the system architecture see Figure 2.1 which represent highly simplified diagram of the overall system.

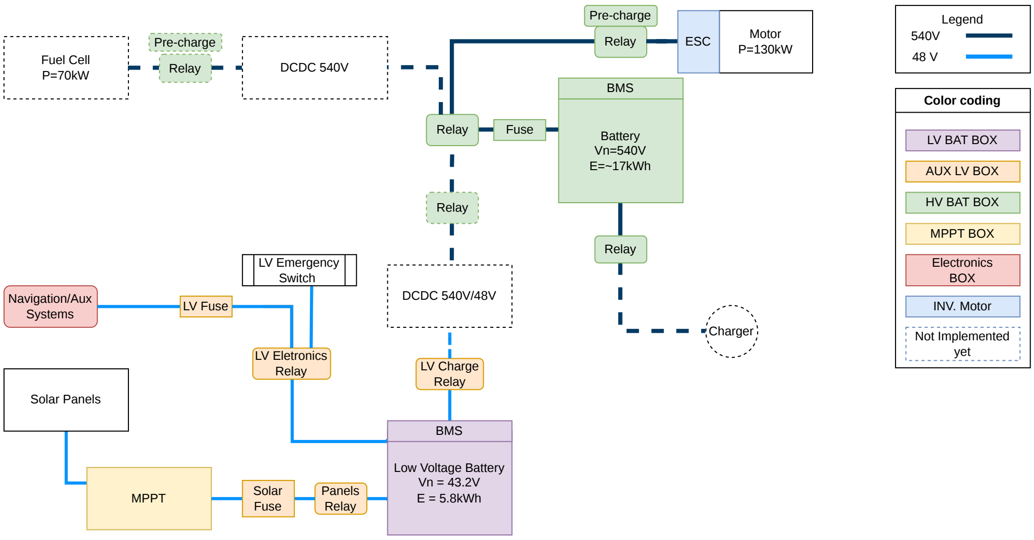


Figure 2.1: Simplified electrical schematic of SG01.

2.2 PV system

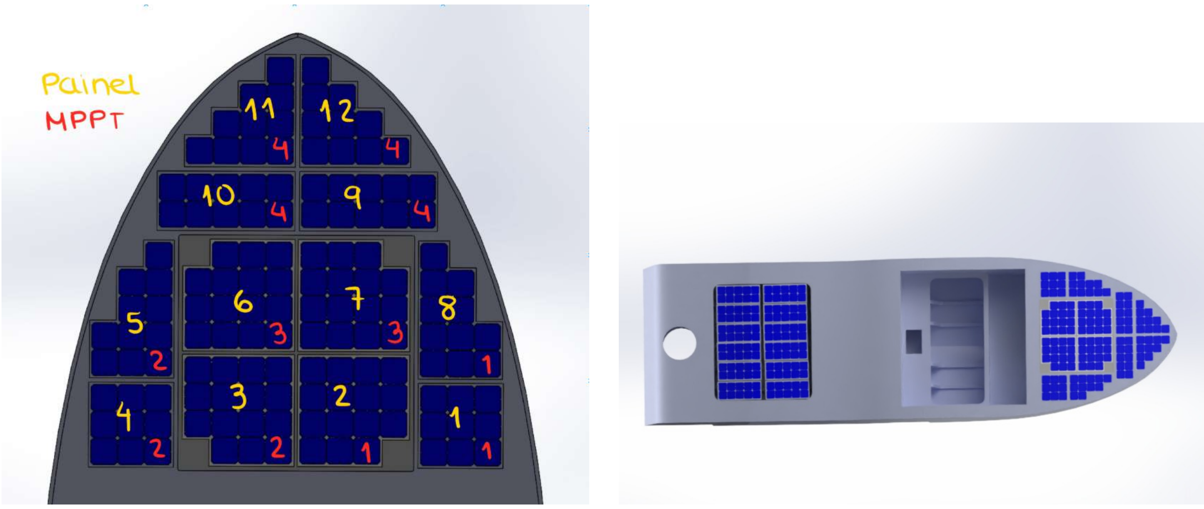
The photovoltaic system of SG01 was designed to maximize the energy harvested from the available deck area while maintaining reliable operation under the varying environmental conditions encountered in marine applications.

The current installation consists of 140 Maxeon Gen 3 (Me3) photovoltaic cells distributed among ten custom-manufactured solar panels. These panels are connected to four independent MPPTs, as illustrated in Figure 2.2a. The distribution of panels among the MPPTs was carefully selected to reduce the effects of partial shading, temperature gradients, and non-uniform solar irradiance across the vessel surface. By electrically separating regions with different operating conditions, each MPPT can independently track the optimal operating point of its associated panels, increasing the overall energy yield.

Each photovoltaic cell has an active area of approximately $155.06\ \mathrm{cm}^2$, resulting in a total photovoltaic area of $2.171\ \mathrm{m}^2$.

Assuming Standard Testing Conditions (STC) with an irradiance of $1000\ \mathrm{W/m}^2$, the photovoltaic array is capable of delivering approximately $526.4\ \mathrm{W}$ of peak power (theoretical value).

The photovoltaic energy is processed by four commercial Genasun MPPTs, which are grouped in a dedicated MPPT box. The MPPTs convert the variable voltage produced by the solar panels into a stable charging voltage suitable for the low-voltage battery system. This architecture allows each photovoltaic group to operate close to its maximum power point independently, improving energy extraction under non-uniform illumination conditions that are common in maritime environments.



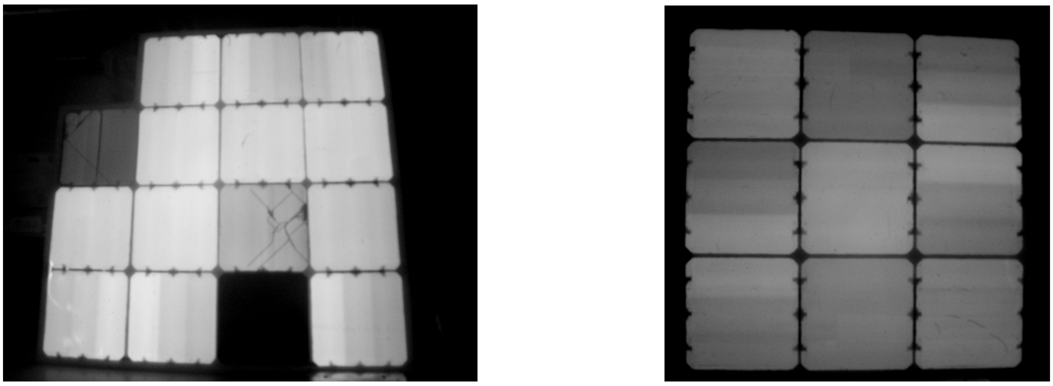
(a) Distribution of Solar Panels and MPPT on SG01.

(b) Render of São Gabriel 01 with additional solar panels.

Figure 2.2: Solar panels installation on SG01.

To ensure long-term reliability, the photovoltaic modules were manufactured in house using materials selected specifically for marine operation.

Prior to installation, all panels underwent electroluminescence testing to detect manufacturing defects such as microcracks, broken cells, or interconnection failures, Figure 2.3. These tests were performed before integration into the vessel to ensure that only panels with acceptable electrical and structural integrity were installed. This quality-control procedure reduces the likelihood of premature degradation and enables comparison before and after competitions.



(a) Worst results.

(b) Best results.

Figure 2.3: Electroluminescence test results.

Although the current photovoltaic installation already provides a significant contribution to the low-voltage energy budget, additional usable deck area remains available. Figure 2.2b presents a conceptual rendering with supplementary solar panels. Preliminary estimates indicate that the total photovoltaic power could be increased to approximately $977\ \mathrm{W}$, almost doubling the currently installed capacity. Such an upgrade would substantially reduce battery charging times and significantly increase the operational autonomy of the vessel.

2.3 Battery

The current low voltage battery uses Samsung INR21700-50G cells in a 12s28p configuration. This configuration results in the specifications written in Table 2.1.

Table 2.1: SG01 and São Rafael 03 (SR03) battery specs.

Parameter	Unit	SG01 Battery	SR03 Battery
Max Voltage	V	50.4	50.4
Nominal Voltage	V	43.2	43.2
Min Voltage	V	30	30
Configuration	–	12s28p	12s7p
Number of cells	–	336	84
Energy Capacity	Wh	5855.56	1512
Capacity	Ah	140	35
Max charge current	A	135.8	42
Cutoff charge current	A	3.395	1.75
Max discharge current	A	271.6	175
Weight	kg	25.2	?
Internal resistance	mΩ	6	24

SG01 battery was designed for a different vessel which needed a substantial high capacity. The battery is too heavy, and overall overdimensioned.

For lab tests and for a more realistic impact analysis, SR03 battery will be used. This is a much smaller battery with a reasonable capacity for this vessel. It uses Samsung INR21700-50s cells in a 12s7p configuration. This configuration results in the specifications also written in Table 2.1.

2.4 PV System Impact on Autonomy

With SG01 battery (5844.56 Wh) and the maximum power that the solar panels can currently provide (526.4 W), the minimum time to fully charge the battery is 11h and 7min (5844.56 Wh / 526.4 W). This is a large time to charge a battery. There are two solutions to this problem: increase the power provided by the solar panels or decrease the battery capacity.

Let's consider the case where we reduced the battery capacity of 1512 Wh which is the capacity of our solar boat (SR03) battery. With this change alone the battery is fully charged in 2h and 53min, which is a much more reasonable time. Additionally, if we increase the power provided by the solar panels to 977 W, the battery will be fully charged in 1h and 33min, which is an excellent time to charge a battery.

The charging times are a good to evaluate the Photovoltaic (PV) system, but the main goal is to maximize the time on water before full battery discharge. For that lets consider 1000 W of consumption. Considering the original battery it takes 5 h 51 min to fully discharge without solar panels and 12 h 21min with solar panels and approximately 10 days with the additional solar panels. As previously said, this battery is overdimensioned and will be changed in the future.

To get a better idea of the impact of the PV system lets consider the battery of SR03. This battery takes 1.51 h to fully discharge without solar panels and 3 h 12 min with solar panels and approximately 2 days 18 h with the additional solar panels.

By reducing the battery capacity and increase the solar panel area, the battery would never need external charging. In another point of view, the low voltage system can drain more power from the battery.

2.5 Environmental changes

In engineer, the word "fast", used in the title of this thesis, needs to be associated with a unit. It is only classified as fast compared to a slower unit. In this section, a brief study was made about waves frequency, boat oscillations and the effect on MPPT algorithm speed.

Tests in Cascais at the marina: I collected 3 datasets—1 while moored at the marina, 2 while underway, and 3 while unmoored at the marina.

I think the wave frequency wasn't the dominant factor because the amplitude is very low, so the boat was oscillating at its natural frequency. I need to calculate the natural frequency to confirm this

It's not the only visible frequency. The rest of the spectrum could be due to: tiny wavelets and wake reflections, mooring line forces, wind gusts, and water motion reflected from seawalls and neighboring boats.

If I'm wrong, the data might be correct and simply show a frequency of 0.5 Hz (higher than the expected 0.05 to 0.3) because we're very close to the coast

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3 State-of-the-Art

In this chapter, the main State-of-the-Art concepts regarding the engineering behind an MPPT charger and solar panels are explained. More specifically, it addresses key challenges such as: what is a solar panel, MPPT algorithms, DC-DC converter types, and battery charging techniques.

3.1 Solar Panels

Photovoltaic (PV) panels are devices that convert solar irradiance into electrical energy. A PV panel consists of multiple solar cells electrically connected in series and/or parallel to achieve the desired voltage and current levels. As in conventional electrical systems, the interconnection topology directly affects the electrical characteristics of the panel: series connections increase the output voltage, while parallel connections increase the output current [15].

Series-connected PV panels operate at higher voltage levels, which reduces conduction losses in interconnecting cables and improves the efficiency of the DC-DC conversion stage. Moreover, power converters designed for high voltage and low current operation are generally less complex and more cost-effective than those operating at low voltage and high current. However, series configurations are more sensitive to partial shading or panel failure, as the output current of the entire string is limited by the weakest module [16]. In contrast, parallel configurations provide improved tolerance to shading and faults but result in higher current levels and increased conduction losses. In this project, series/parallel configuration of the solar panels will not be changed, since it is already pre-defined by Técnico Solar Boat (TSB), but it is important to understand the implications of each configuration.

Each cell is made of semiconductor materials, usually silicon. This semiconductor is doped with phosphorus, a group V element, to create a negative type layer. On the other side, a layer is doped with boron, a group III element, to create a positive type layer. This creates a p-n junction, which is essential for the photovoltaic effect [2].

When sunlight arrives at the solar cell, the photons energy is absorbed by the semiconductor material. This energy excites electrons, allowing them to escape their atomic bonds and create electron hole pairs. The electric field at the p-n junction drives these free electrons towards the n-type layer and holes towards the p-type layer, generating a flow of electric current when the cell is connected to an external circuit [2], Figure 3.1.

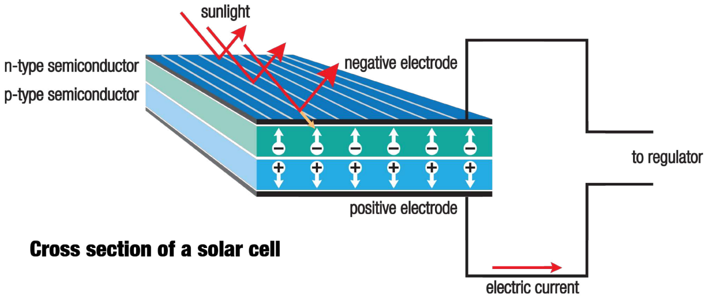


Figure 3.1: Principle of operation of a solar panel [1].

The energy generated results in voltage and current outputs that are not constant. Voltage and current are interdependent, therefore, a variation in one parameter leads to a corresponding variation in the other. This relationship is nonlinear and can be characterized by a current-voltage (I-V) curve. Furthermore, the power output also varies with operating conditions and can be represented by a power-voltage (P-V) curve. Both curves are represented in Figure 3.2. In both curves, there is a specific point where the power output is maximized, known as the Maximum Power Point (MPP). This point corresponds to a unique combination of voltage and current, denoted as V_{mpp} and I_{mpp}, respectively. Operating the solar panel at this point ensures optimal energy extraction under a given condition.

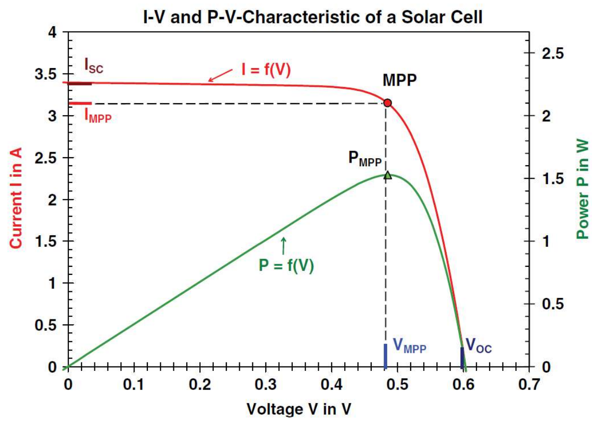
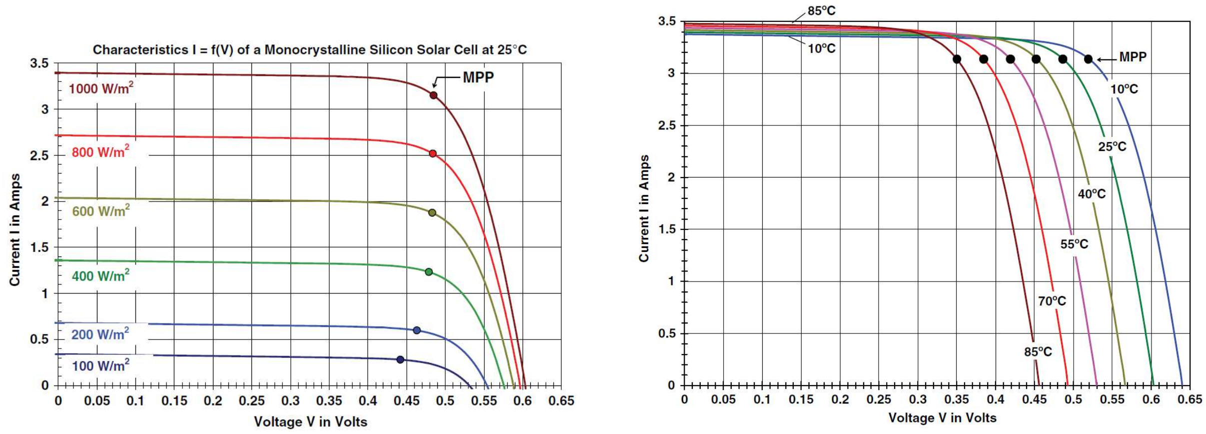


Figure 3.2: I-V and P-V curves of a solar panel [2].

The characteristic curves of a solar panel depend on various factors, including the quality of the materials used, the solar cells, and environmental conditions such as temperature, solar irradiance, wind, and shades. It is noted that under partial shading conditions, it is possible to have multiple local maxima, but overall, there is still only one true MPP [17].

As seen in Figure 3.3a, the variation of the solar irradiance changes the I-V and P-V curves of the solar panel. The increase in irradiance produces more power by mainly increasing the current output of the panel. On the other hand, the temperature has an opposite effect (Figure 3.3b). The increase in temperature produces a decrease in power output, mainly by decreasing the voltage output of the panel.



(a) Cell temperature of 25 ºC and variable Irradiance.

(b) Irradiance of 1 $\mathrm{kW/m^2}$ and variable temperature.

Figure 3.3: I-V curves under different conditions [2].

Both environmental conditions have a meaningful impact on the MPP of the solar panel and therefore on the power output of the system. To maximize the power extracted by the solar panels, a classic power converter, like a buck converter with fixed duty cycle, will not be efficient since neither the current nor the voltage are constant for different conditions. Instead, an MPPT charger is used, which will find the MPP and operate the solar panel at that point while sensing the changes caused by the environment.

3.2 Maximum Power Point Tracker (MPPT)

A Maximum Power Point Tracker (MPPT) charger is a device that is used to optimize the power output from solar panels by continuously tracking and adjusting the operation point of the panels to ensure they operate at their MPP.

To achieve its goal, an MPPT charger is usually composed of a DC-DC converter, a microcontroller, and some sensors (Figure 3.4). The sensors are used to measure the input/output variables of the control system (typically voltage and current). The information acquired by these sensors is processed by the microcontroller, which runs an algorithm to determine if the MPP was reached or how to act on the system towards the MPP. In the second case, the microcontroller generates a signal to control the DC-DC converter that will adjust its operation accordingly.

The output can be connected to different types of loads, like batteries, DC loads or inverters. This project focuses on batter charging applications.

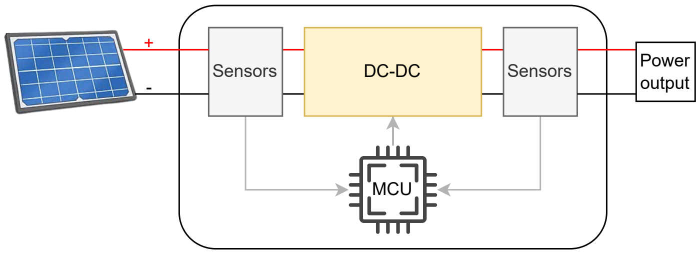


Figure 3.4: Simplified structure of an MPPT charger.

3.3 MPPT Algorithms

There are plenty of MPPT algorithms, each one with its own advantages and disadvantages. The choice of the algorithm will depend on the specific application, the desired performance, and the available resources. In this work, the aim is to extract the maximum energy possible of a solar panel installed in a moving boat, which will produce an I-V curve variation due to the quick changes in irradiance caused by changes of inclination, orientation and clouds, and also temperature changes due to waves and wind. For these reasons, tracking speed and accuracy are the most important factors to consider.

With this in mind, the most relevant algorithms are briefly reviewed in order to identify the most suitable option for this project.

3.3.1 Constant Voltage (CV)

This is the simplest and most inefficient method. It works by operating the solar panel at a fixed voltage. This referenced voltage is used to calculate the duty cycle of the DC-DC converter that will maintain the operating point of the solar panel.

Well-known variants improve performance by setting the reference voltage to a fraction of the open-circuit voltage (typically 71-78%) [17]. This voltage is obtained by briefly disconnecting the panel from the load, increasing reference accuracy and efficiency. Another variant uses a fraction of the short-circuit current as the reference [18].

Although both methods are simple and cost-effective, they are inherently inefficient due to the necessity of interrupting energy production during measurement phases. However, alternative implementations address this limitation through proxy measurement techniques employing dummy cells or diodes, whose physical properties closely approximate those of standard solar cells, thereby enabling continuous power generation during measurement acquisition [17] [19]. Still, the MPP is not always located at the same percentage of the open-circuit voltage or short circuit current, leading to a lack of accuracy and therefore lower efficiency [14].

3.3.2 Perturb and Observe (P&O)

The Perturb and Observe (P&O) method is widely used in commercial products and is the basis of many advanced algorithms. Its popularity lies in its simplicity, low cost, and ease of implementation.

As its name indicates, the algorithm perturbs the voltage of a PV array and observes the resulting effect on the output power [20]. If an increase in voltage leads to an increase in output power, the operating point is moving toward the MPP, and the algorithm continues to increase the voltage. Conversely, if the output power decreases, the operating point is moving away from the MPP, and the direction of the perturbation is reversed [3, 21]. Figure 3.5 illustrates the flowchart of the algorithm.

The biggest drawback of this algorithm is that it oscillates around the MPP, which produces power losses. This oscillation can be reduced by decreasing the step size of the perturbation, but this will also reduce the tracking speed of the algorithm. This results in an inherent trade-off between tracking speed and accuracy [17, 18].

To mitigate this issue, an adaptive step size can be used, where the step size is larger when the MPP is far away and smaller when it is closer. Consequently, the tracking speed is maximized while minimizing the oscillations around the MPP. This can be achieved by measuring the power and voltage and calculating the P-V curve slope ($\Delta P/\Delta V$). At the MPP, this slope approaches zero, allowing the step size to be reduced accordingly. This adaptive variant is commonly referred to as the differential power perturb-and-observe(dP-P&O)".

Another weal-know issue of P&O is that it can get confused in rapidly changing environmental conditions, like fast irradiance changes caused by clouds. In this case, the algorithm can misinterpret the power change caused by the environmental variation as a result of its own perturbation, leading it to move away from the MPP instead of towards it [17]. For example, if the perturbation was in the wrong direction, but the irradiance increased, the power output would increase, and the algorithm would continue perturbing in the wrong direction (Figure 3.6).

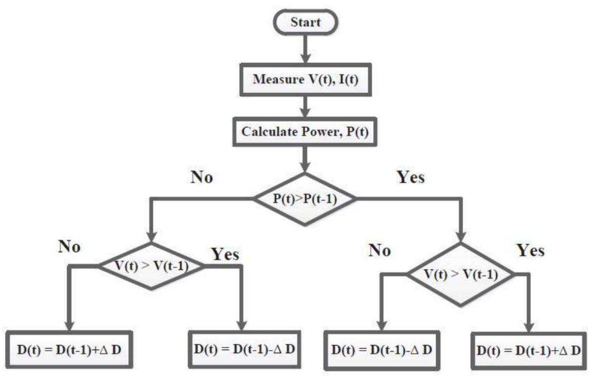


Figure 3.5: Flow chart of the classic version of Perturb and Observe algorithm [3].

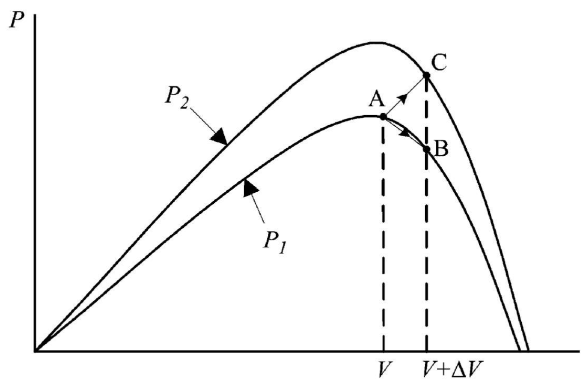


Figure 3.6: Divergence of P&O from MPP as shown in [4].

To address this limitation, an additional condition is incorporated into the algorithm that evaluates two consecutive measurements of ΔP and ΔV. Specifically, when the signs of consecutive ΔP measurements do not follow the signs of ΔV, this indicates that the observed power variation results from environmental disturbances rather than the algorithm's perturbation. Consequently, the voltage adjustment is suppressed. This refined variant is designated the two-point algorithm or improved P&O method [15].

There are other variations of P&O algorithm that improve its performance, like Variable Step Size P&O (VSS-P&O), the three-point P&O, a-factor P&O, and more. Unfortunately, due to the scope of this work, it is not possible to explain all of them.

3.3.3 Incremental Conductance (IncCond)

The Incremental Conductance (IncCond) algorithm is another widely used MPPT method due to its accuracy and ability to track the MPP under rapidly changing environmental conditions.

This algorithm is based on the fact that at the MPP, the derivative of power with respect to voltage is zero. By knowing the output voltage and current of the solar panel, the algorithm can calculate the conductance and the incremental conductance [22].

The following formulas, Equation 3.1, show the equation on which this algorithm is based.

$$\frac{dP}{dV} = \frac{d(V \cdot I)}{dV} = I + V\frac{dI}{dV} \rightarrow \frac{1}{V} \times \frac{dP}{dV} = \frac{I}{V} + \frac{dI}{dV} \tag{3.1}$$

At the MPP point, the slope of the P-V curve is zero, which means that the negative of the conductance, $-I/V$, is equal to the incremental conductance, dI/dV, (Figure 3.8). The algorithm compares these two values to determine if the operating point is at, to the left, or to the right of the MPP [18] [17]:

$$\begin{aligned} \frac{dP}{dV} = 0 &\iff \frac{dI}{dV} = -\frac{I}{V} && \text{at MPP} \\ \frac{dP}{dV} > 0 &\iff \frac{dI}{dV} > -\frac{I}{V} && \text{left of MPP} \\ \frac{dP}{dV} < 0 &\iff \frac{dI}{dV} < -\frac{I}{V} && \text{right of MPP} \end{aligned} \tag{3.2}$$

Based on these conditions, the algorithm adjust its operating point (increasing or decreasing the voltage) using a variable step size based on the instantaneous conductance relative to the incremental conductance, Figure 3.7. For example, as suggested in [17], the step size can be calculated using a proportional integral (PI) controller with zero as input reference and error defined as:

$$e = \frac{I}{V} + \frac{dI}{dV} \tag{3.3}$$

Comparative analysis demonstrates that the IncCond algorithm exhibits superior performance relative to the P&O method. This outcome is expected, as the Incremental Conductance technique was developed specifically to address the limitations inherent in the P&O approach [22]. For example, the IncCond method eliminates the oscillation problem around the MPP that is characteristic of the P&O method. Additionally, IncCond demonstrates superior performance under rapidly changing environmental conditions, such as sudden irradiance variations, since it uses derivative-based information that is less sensitive to transient disturbances [15].

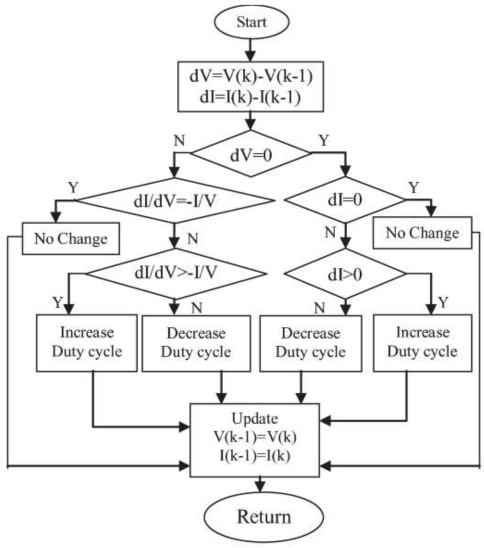


Figure 3.7: Flowchart of the IncCond method with direct control [5].

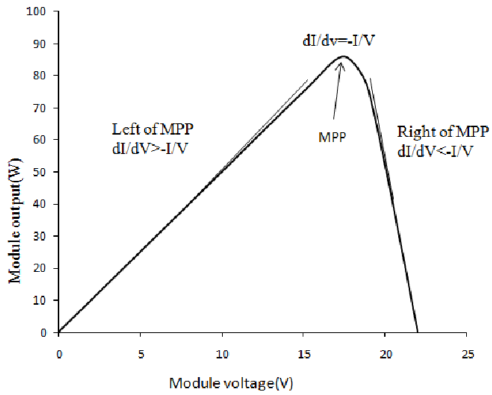


Figure 3.8: IncCond method principle.

3.3.4 Look Up Table (LUT)

The Look-Up Table (LUT) method is a simple and fast MPPT algorithm that relies on pre-calculated/measured data to determine the optimal operating point of a solar panel [18].

The LUT table contains several entries organized by voltage and current. Each of these entries contains the optimal duty cycle of the DC-DC converter, that are pre-calculated or measured for a specific voltage and current. This approach allows the algorithm to reach the MPP in one clock cycle by measuring the voltage and current of the solar panel, looking for the closest entry in the LUT table, and applying the corresponding duty cycle to the converter. The entries of the table can be calculated through a mathematical model of the solar panel with different conditions (temperature and irradiance) or measured in a laboratory.

Table 3.1 is a visual representation of a 2D LUT, only based on current and voltage. Each entry represents the duty cycle used in each case.

The main advantage of this method is its speed since it can reach the MPP in one clock cycle. This makes it suitable for applications where fast tracking is required, like in this project. However, it is not very accurate [3].

Table 3.1: LUT matrix representation with $m \times n$ elements.

	V_1	$\cdots$	V_k	$\cdots$	V_m
I_1	D_{11}	$\cdots$	D_{1k}	$\cdots$	D_{1m}
$\vdots$	$\vdots$		$\vdots$		$\vdots$
I_j	D_{j1}	$\cdots$	D_{jk}	$\cdots$	D_{jm}
$\vdots$	$\vdots$		$\vdots$		$\vdots$
I_n	D_{n1}	$\cdots$	D_{nk}	$\cdots$	D_{nm}

To improve the accuracy, the LUT table needs to be larger, which increases the required memory resources. Also, by adding the temperature and/or irradiance to the table would increase the accuracy, making it almost 100% accurate, but this would increase the size of the table exponentially. In extreme cases, the memory requirements became expensive and slow. There is a trade-off between size, speed, and accuracy that needs to be considered when using this method. Also, the LUT method does not adapt to the aging of the solar panel or changes in its characteristics over time, unless the table is updated periodically through new measurements.

3.4 Types of DC-DC Converters

The DC-DC converter is a crucial component of an MPPT charger, as it allows the adjustment of the operating point of the solar panel to match the MPP. His choice is highly important to the project since it directly affects the efficiency, cost, and complexity of the overall system.

DC-DC converters can be classified into two main categories: isolated and non-isolated converters. The main difference between these two types of converters is the presence or absence of galvanic isolation between the input and output circuits [23].

3.4.1 Non-isolated versus isolated DC-DC converters

This galvanic isolation is usually achieved by using a transformer, which provides electrical separation between the input and output sides of the converter. This isolation is important in applications where safety is a concern, such as in medical devices or industrial equipment, as it helps to prevent electrical shock and damage to sensitive components. In the context of solar energy systems and E-mobility, isolated converters are often used when the solar panel is connected to the grid [24], as they help to protect against voltage spikes and other electrical disturbances. Also, non-isolated converter can provide very higher/lower gain, which is useful for applications where solar panels MPP voltage is very different from the battery or load voltage (typically medium/high voltages which is not the case).

In contrast, non-isolated converters do not provide galvanic isolation between the input and output circuits. However, they generally offer higher efficiency, lower cost, reduced size, and lower circuit complexity (Table 3.2). Additionally, they present fewer challenges in thermal management and are simpler to design. These advantages, together with the fact that galvanic isolation is not a requirement for this application, led to the exclusion of isolated DC-DC converters from this project.

Table 3.2: Comparison between Non-isolated and isolated converters.

Type	Non-isolated Converters	Galvanic Isolated Converters
Circuit Complexity	Low	High
Efficiency	High	Medium
Size	Compact	Large
Electromagnetic Interference (EMI)	Medium/High	High
Switching Frequency	Medium	Medium
Thermal Management	Easier	Moderate
Control Complexity	Moderate	Moderate
Gain	Near 1	Very High/low
Cost	Low	High

Legend: undesired medium good

3.4.2 Non-isolated DC-DC converters

Modern non-isolated conversion circuits generally use one of three basic topologies: buck, boost, or buck-boost converters. They are "basic" in the sense that only one switching element is needed. A given topology is used to obtain a specific result, such as voltage step-down, voltage step-up, or hybrid mode [7].

The **boost converter** is a step-up DC-DC converter widely employed in MPPT controllers when the required output voltage is higher than the solar panel open circuit voltage. It uses an inductor to store energy when the transistor is in the ON state and the diode is reverse-biased. When the transistor is on the OFF state, the inductor releases the stored energy to the output through the diode, increasing the output voltage (Figure 3.9b) [25].

This topology has been extensively documented in the literature and is distinguished by its inherent simplicity, cost-effectiveness, and superior efficiency characteristics [12] [3] [22]. Depending on the design variation used and constraints, it can achieve conversion efficiencies up to 98% [26].

A limitation of this topology is that it cannot emulate an impedance lower than the load impedance, which prevents it from reaching values close to the short-circuit current of the PV module. Consequently, it is not compatible with all solar panels or MPPT algorithms [26, 27].

The **buck converter** is a step-down DC-DC converter also widely used for applications where the solar panel voltage is higher than the battery or load voltage. Similar to the boost converter, it uses an inductor to store energy when the transistor is in the ON state and the diode is reverse-biased. However, in this case, the input is connected to the output when the transistor is in the ON state. When the transistor is in the OFF state, the inductor releases the stored energy to the output through the diode, decreasing the output voltage (Figure 3.9a) [26].

This converter also has a similar downside to the boost converter. The buck converter can not emulate a smaller impedance than the load impedance, and therefore, it can not reach values near the open circuit voltage of the PV module [26] [27].

Buck-Boost converters are step-up and step-down DC-DC converters that can operate in both modes. This makes them more versatile than the previous two topologies, but also more complex and less efficient [26]. This converter uses an inductor to store energy when the transistor is in the ON state and the diode is reverse-biased. When the transistor is in the OFF state, the inductor releases the stored energy to the output through the diode, increasing or decreasing the output voltage (Figure 3.9c) [7].

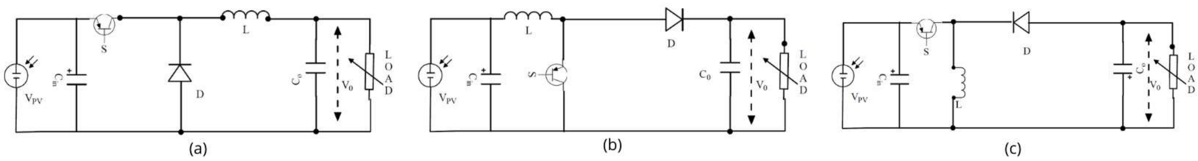


Figure 3.9: Buck (a), Boost (b) and Buck-Boost (c) DC-DC converters.

The conventional buck-boost converter produces an output voltage inverted with respect to the input which creates challenges on sensor measurements, driving circuit and isolation, consequently, it is unsuitable for the battery charging system [25].

An enhanced variant of the buck-boost converter is the CUK converter, illustrated in Figure 3.10a. The main characteristic of the CUK converter is that the currents through inductors L_1 and L_2 are equal, allowing the use of a common magnetic core, which helps reduce current ripple. However, a major drawback of this topology is that the entire load current must pass through the coupling capacitor C. In addition, the converter produces an inverted output voltage [25].

The issue of output voltage polarity inversion can be addressed by using a single-ended primary inductor converter (SEPIC), as shown in Figure 3.10b. This makes the SEPIC converter suitable for battery charging applications. Nevertheless, its main drawbacks are the high output current and voltage ripple which increases losses and electromagnetic interference (EMI) [28]. These limitations are mitigated by the ZETA converter, shown in Figure 3.10c, which provides improved output current characteristics while maintaining the same output voltage polarity as the SEPIC converter [21] [25].

3.4.3 Comparison between Non-isolated DC-DC converters

All the non-isolated DC-DC topologies explained before have their own advantages and disadvantages. The choice of the topology will depend on the specific application, the desired performance, and the available resources. To help with this choice, Table 3.3 shows a comparison between the most relevant

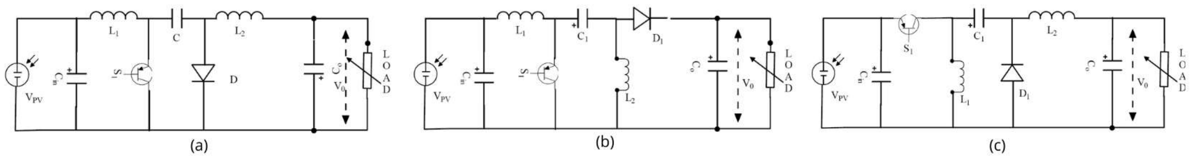


Figure 3.10: CUK (a), SEPIC (b) and ZETA (c) DC-DC converters.

characteristics of each topology [28]. The most significant advantages of each topology are highlighted accordingly with the aim of the project.

Table 3.3: Comparison of non-isolated DC-DC topologies.

Topology	Boost	Buck	Buck-Boost	ZETA	CUK	SEPIC
Electrical performance						
Efficiency	High	High	High	Medium/High	Medium	Medium
EMI	Low	Low	High	Medium	Medium	Medium
Input current ripple	Low	High	High	Low	Very low	Medium
Output current ripple	High	Low	High	Very low	Low	Medium
Output voltage ripple	Medium	Low	Medium	Low	Medium	High
Structural characteristics						
Circuit complexity	Low	Low	Low	Medium	Medium	Medium
Size	Low	Low	Low	Medium	Medium	Medium
Storage elements	1 L	1 L	1 L	2 L, 1 C	2 L, 1 C	2 L, 1 C
Switch stress	No	Yes	No	Yes	Yes	Yes
Functional aspects						
Type	Step-Up	Step-Down	Step-Down/Up	Step-Down/Up	Step-Down/Up	Step-Down/Up
Versatility	Low	Low	Medium	High	Medium	High
Inverted output	No	No	Yes	No	Yes	No
Reliability	High	High	High	Medium	Medium	Medium
Economic						
Cost	Low	Low	Low	Medium	Medium	Medium

Legend: good important

For MPPT controllers, the choice is mainly governed by the possibility of operating at the MPP which is connected with the converter type (Step Up, Down or Down/Up). Also, efficiency, cost (related to the number of storage elements), output voltage ripple are relevant in this choice.

The buck converter can be excluded from this comparison since in this project the solar panel voltage is lower than the battery voltage, therefore, a step-down converter is not suitable. However, when choosing between the boost and the step-up/down converters, there is a trade-off. The boost converter is simpler, cheaper and more efficient. On the other hand the step-down/up converters are more versatile since they can operate in a wide range of input voltages, which is interesting since the size of the solar panels and battery voltage can change.

Considering that in this project the solar panel voltage is always lower than the battery voltage, the boost converter is the most suitable topology due to its simplicity, cost-effectiveness, and high efficiency. However, if versatility were to become a priority, the ZETA converter would be the preferred choice among the step-up/down topologies, owing to its non-inverted output voltage and low output current ripple, which are advantageous for battery charging applications.

3.5 Battery charging techniques

As the demand for electronic devices and E-vehicles increases, the need for efficient, compact, and lightweight batteries has emerged. Among many existing technologies, lithium-ion batteries have one of the best energy-to-weight/volume ratios and, at this moment, it is the technology used in TSB batteries. But, like every other battery technology, they need to be charged properly to ensure their safety and longevity [6].

To charge these batteries properly, the MPPT algorithm can operate alone if an additional converter is used in series. But in most of MPPT chargers, the MPPT and the battery charging unit are integrated in the same converter. This way, the MPPT algorithm can adjust the operating point of the solar panel to extract the maximum power while the battery charging technique ensures that the battery is charged properly. A proper combination of both algorithms/techniques is essential to ensure both maximum energy extraction and battery safety.

There are several techniques to charge lithium-ion batteries, the most relevant for this use case being the constant current-constant voltage (CC-CV) method [29]. Also, it is widely used in commercial chargers due to its simplicity and effectiveness [30]. This method consists of two main phases (Figure 3.11). In the first phase, the battery is charged with a constant current until it reaches a predefined voltage limit (usually 4.2 V per cell). In the second phase, the voltage is held constant at this limit while the current gradually decreases as the battery approaches full charge. Once the current drops below a certain threshold, the charging process is terminated to prevent overcharging [30].

When employing the CC-CV charging technique with a MPPT controller, the initial Constant Current (CC) phase is regulated by the MPPT algorithm, which extracts the maximum available power from the solar panel limited by the maximum charging current. Once the Constant Voltage (CV) phase is reached, the MPPT function is disabled, and the power converter increases the solar panel voltage to progressively reduce the charging current while keeping the output voltage constant.

More advanced high-performance battery charging techniques have been proposed in literature to reduce charging time without increasing the charging current, thereby preserving battery lifetime. However, these techniques generally introduce higher switching losses and increased system complexity. In the present application, the current generated by the photovoltaic panels is less than half of the maximum allowable battery charging current, consequently, the charging current is inherently limited by the solar panels rather than by the charging strategy. Furthermore, battery lifetime is not a critical concern in this project, as the battery is expected to undergo a limited number of charge cycles, not exceeding 20 cycles per year.

Therefore, the CC-CV charging method is considered the most suitable technique for this project,

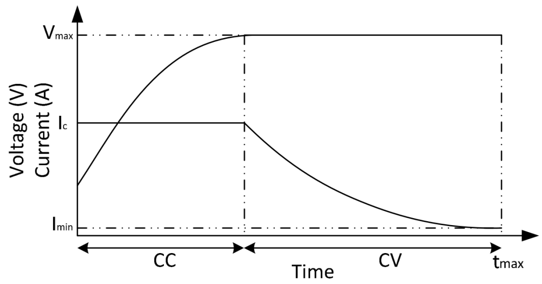


Figure 3.11: Charging profile of CC/CV [6].

owing to its simplicity, effectiveness, and widespread adoption in commercial battery chargers.

Other charging techniques fall outside the scope of this work, however, some methods are briefly discussed in the references provided [29] [6] [30].

3.6 Commercial Options

To assess the feasibility of adopting a Off the shelf (OTS) solution, this section reviews existing commercial options for solar battery charging with MPPT capability. Both specialized Integrated Circuits (ICs) and complete commercial MPPT chargers are considered, and their electrical, functional, and economic characteristics are evaluated against the requirements of this project. The objective is to identify whether a commercially available solution can satisfy the constraints of the application or, alternatively, to justify the need for a custom-designed converter and control system.

3.6.1 Specialized ICs

Several commercially available Integrated Circuit (IC) are specifically designed for solar battery charging and MPPT applications. These devices are relevant to this project since, if suitable, they could significantly reduce design complexity by integrating functions such as DC-DC conversion, protection mechanisms, and MPPT algorithm.

Table 3.4 summarizes four commonly used commercial ICs. Characteristics marked in red indicate incompatibilities with the project requirements, leading to the exclusion of the corresponding devices.

The BQ24650RVAT is a Stand-Alone Synchronous Buck Battery Charge Controller which implements a high-efficiency battery charger while also providing safety features. By adding a Microcontroller Unit (MCU) to run a MPPT algorithm and current and voltage sensors, this IC can track the MPP of a solar panel, making it appropriate for applications with 12/24 V batteries and high-power panels, such as electric vehicles. However, it does not support the required 48 V battery voltage and is therefore unsuitable for this work.

The MAX20801TPBA+ and SPV1040TTR target low-power IoT applications with small batteries and solar panels. Both devices are unable to meet the 48 V output requirement, and their limited publicly available information on tracking speed further restricts performance evaluation.

The LT8491IUKJ#PBF offers a wide input and output voltage range, buck-boost operation, high efficiency, communication capabilities, and CC-CV charging. Despite these advantages, its reported MPPT tracking speed of up to 2 s is insufficient for the dynamic conditions of this application. Additionally, its I^2C interface would require an additional microcontroller to provide CAN communication, increasing system cost.

Table 3.4: Comparison of commercial specialized ICs.

Model	BQ24650RVAT	MAX20801TPBA+	SPV1040TTR	LT8491IUKJ#PBF
Electrical specifications				
Input voltage	5 to 28 V	1.5 to 18 V	0.3 to 5.5 V	6 to 80 V
Output voltage	12 to 24 V	12.4 V	-0.3 to 5.5 V	1.3 to 80 V
Max output current	Externally limited	12 A	1.8 A	10 A
Switching frequency	600 kHz	Unknown	100 kHz	100 to 400 kHz
Electrical efficiency	95%	99.1% (max)	80 to 95%	95 to 99%
MPPT and control				
Algorithm	User input	Unknown	Unknown	P&O
Tracking accuracy	User dependent	99.9% (max)	Unknown	Unknown
Tracking speed	User dependent	Unknown	Unknown	1 to 2 s
Functional aspects				
Topology	Synchronous Buck	Synchronous Buck	Synchronous Boost	Buck-Boost
Communication	No	No	No	I^2C
Configurability	No	No	No	Yes
Charge profile	CC-CV	Unknown	Unknown	CC-CV
Extra hardware needed				
Transistors	Yes	No	No	Yes
Sensors	Yes	No	No	No
Storage elements	Yes	Yes	Yes	Yes
Economic				
Price	5.25 €	4.50 €	3.02 €	15-20 €

Legend: unsuitable

3.6.2 Commercial MPPT chargers

Several commercial MPPT chargers are available that can address the problem under analysis without requiring additional hardware. However, each solution presents limitations when evaluated against the project requirements. Table 3.5 summarizes selected commercial MPPT chargers, with red-marked characteristics indicating incompatibilities.

The GVB-8-Li-CV(50.4) is a high-performance and fast MPPT charger, particularly suited for dynamic environments such as marine applications. It has been successfully used by TSB in previous projects, demonstrating high efficiency and reliability. Nevertheless, its lack of configurability, fixed output voltage, and high cost restrict its suitability for this application. Despite these drawbacks, it serves as a valuable reference design.

The Smart Solar MPPT 100/20, developed by Victron Energy, offers high efficiency, fast tracking, and integrated communication interfaces. However, it uses a buck topology that requires higher input voltages, making it incompatible with the small and distributed solar panel arrays used in the TSB application.

Other commercial solutions, such as the Rover Lite and MS4840N, provide acceptable performance but rely on communication protocols that are not compatible with the project requirements and offer limited technical documentation.

The Reboost V0.2.1 is an open source MPPT charger originally developed for automotive applications. While it provides flexibility and near adequate specifications, its reliance on expensive and discontinued components, along with the use of a boost topology, limits scalability and adaptability to potential future battery voltage changes.

Overall, none of the evaluated commercial MPPT chargers fully satisfy the technical, economic, and configurability requirements of this project, motivating the development of a custom solution.

Table 3.5: Comparison of commercial MPPT controllers.

Model	**GVB-8-Li-CV (50.4V)**	**Smart Solar MPPT 100/20**	**Reboost V0.2.1**	**Rover Lite**	**MS4840N**
General information					
Brand	Gensun	Victron Energy	TPEE	Renogy	BougeRV
Type	Boost	Buck	Synchronous Boost	Boost	Boost
Control and communication					
Communication	No	VE.can and Bluetooth	Yes	Bluetooth and RS485	Bluetooth
Configurable	No	Yes	Yes	Yes	Yes
GUI	No	Yes	Yes	Yes	Yes
MPPT and charging					
Prog. battery voltage	No	Yes	Yes	non-lithium batteries	Yes
Tracking speed	15 Hz / 66.6(6) ms	Fast (unknown)	Programmable	Unknown	Unknown
Tracking efficiency	99%+ typical	Unknown	Programmable	99%	Unknown
Electrical efficiency	96-99% typical	98% peak	Unknown	97%	Unknown
Charger profile	CC-CV	Unknown	Programmable	Unknown	CC-CV
Hardware implementation					
MCU	ATtiny461A-U	Unknown	STM32G474	Unknown	Unknown
MCU Details	8-bit, RISC, 64 MHz	–	32-bit, Arm, 170 MHz	–	–
Transistor	FZT951	Unknown	GS61008T	Unknown	Unknown
Transistor type	PNP BJT	Unknown	GaN FETs	Unknown	Unknown
Economic					
Price	240 €	100-200 €	Components: 250-280 € PCB: 20-100 € Total: 270-380 €	300-350 €	120-160 €

Legend: unsuitable

4 Implementation

introdução do capítulo

4.1 DC-DC converter

introdução do capítulo

4.1.1 Operation principles

The DC-DC converter chosen for this project was a boost converter. This topology uses one switching device, one inductor, one output capacitor and a diode, Figure 4.1. The switching device is used to control the output voltage by switching on and off at a specified frequency (fsw) and duty cycle (D). The inductor is used to store energy when the switch is on (stage 1) and release it when the switch is off (stage 2), while the output capacitor is used to smooth the output voltage, Figure 4.1.

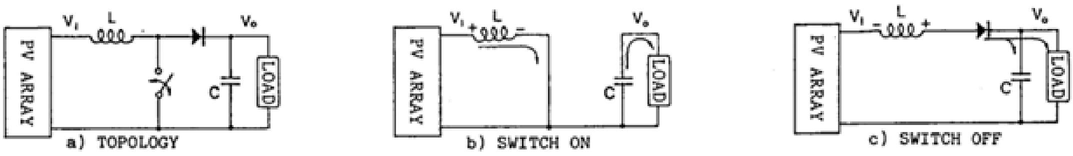


Figure 4.1: Classic Boost converter topology and stages [7].

Analyzing the boost converter separately on his two stages, it is possible to define the drop voltage across the inductor (VL), as a function of the input voltage (Vin) and output voltage (Vout) as follows:

$$V_L = \begin{cases} V_{in} & \text{if } t \in [0, D \cdot T] \\ V_{in} - V_{out} & \text{if } t \in [D \cdot T, T] \end{cases} \tag{4.1}$$

With equation 4.1 and the fact that at stationary state the average voltage across the inductor is zero, it is possible to define the output voltage in function of the input voltage and duty cycle as follows:

$$V_{Lavg} = \frac{1}{T}(\int_0^{DT} V_{in}dt + \int_{DT}^{T}(V_{in} - V_{out})dt) = 0 \Rightarrow V_{out} = \frac{V_{in}}{1-D} \tag{4.2}$$

Equation 4.2 is one of the most important equations in boost converter design, as it defines the relationship between the input voltage, output voltage and duty cycle. As the duty cycle increases, the output voltage also increases, which is the main principle behind boost converters. For our use case, it is import to analyze the relation between the input and output current as well. Considering no power losses, it is possible to define the input current as follows:

$$P_{in} = P_{out} \Leftrightarrow V_{in} \cdot I_{in} = V_{out} \cdot I_{out} \Leftrightarrow I_{in} = \frac{V_{out}}{V_{in}} \cdot I_{out} \Leftrightarrow I_{in} = \frac{I_{out}}{1-D} \tag{4.3}$$

However, these equations are only valid for ideal components since it does not take into account the parasitic elements of the components, such as the resistance of the inductor, the drain source resistance of the transistor, and the voltage drop across the diode (there are more parasitic components that affect the performance). In a real boost converter, these parasitic elements will cause a voltage drop and therefore reduce the output voltage.

Let's quickly analyze the influence of the inductor resistance (R_L) and the voltage drop across the diode (V_d) by changing the equation 4.1 to:

$$V_L = \begin{cases} V_{in} - R_{DS_{(on)}} \cdot I_{DS} & \text{if } t \in [0, D \cdot T] \\ V_{in} - V_{out} - V_d & \text{if } t \in [D \cdot T, T] \end{cases} \tag{4.4}$$

In this case the at stationary state the average voltage across the inductor is $R_L \cdot I_L$ and therefore the output voltage can be defined as:

$$V_{Lavg} = \frac{1}{T}(\int_0^{DT} (V_{in} - R_{DS_{(on)}} \cdot I_{DS})dt + \int_{DT}^{T} (V_{in} - V_{out} - V_d)dt) \Leftrightarrow \tag{4.5}$$

$$\Leftrightarrow R_L \cdot I_L = \frac{1}{T}[T \cdot D \cdot (V_{in} - R_{DS_{(on)}} \cdot I_{DS}) + T \cdot (1-D) \cdot (V_{in} - V_{out} - V_d)] \Leftrightarrow V_{out} = \frac{V_{in} - R_{DS_{(on)}} \cdot I_{DS} - R_L \cdot I_L}{1-D} - V_d \tag{4.6}$$

Equation 4.6 shows the 3 mains causes of losses in a boost converter. The main cause is the voltage drop across the diode, which causes a significant reduction in the output voltage (usually 0.3 V to 0.7 V). This will be solved by using a synchronous boost converter (Figure 4.2), which replaces the diode with a transistor. In synchronous boost converters the voltage drop across the transistor is much smaller due to the low $R_{DS,on}$, usually lower than 0.1 V.

As for the inductor internal resistance, it is not possible to remove it entirely, so it must be taken into account when designing the boost converter. The same is true for the drain source resistance of the transistor, which is also not possible to remove entirely. Therefore, it is important to select components with low internal resistance to minimize the losses and increase the efficiency of the converter.

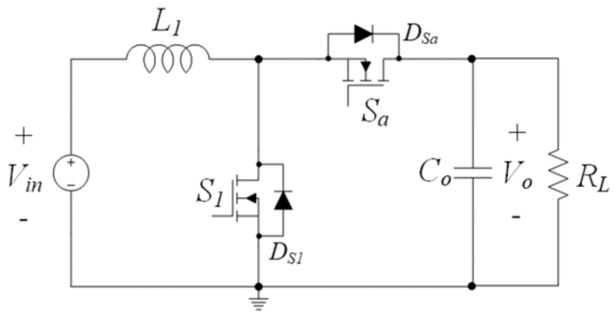


Figure 4.2: Synchronous boost converter topology [8].

4.1.2 Dimensioning

Let us first define the design goals of the synchronous boost converter. The converter must be able to step up the voltage from the solar panels, which may consist of a maximum of 40 cells (Maxeon Gen 3 from SunPower) connected in series, a minimum of 20 cells in series, and is optimized for 35 cells (Figure 4.1). The converter must charge a lithium-ion battery with a nominal voltage of 43.2 V. The switching frequency was defined as 100 kHz. This relatively high frequency allows the use of smaller inductors and capacitors, while still avoiding excessive switching losses.

Table 4.2 and 4.1 summarizes the system requirements.

Table 4.1: PV panel parameters at MPPT.

Configuration	V_{mppt} (V)	I_{mppt} (A)	P_{mppt} (W)
40 cells	24.68	6.05	139.6
35 cells	21.875	6.05	122.15
20 cells	12.34	6.05	69.8

Table 4.2: Converter design constraints.

Parameter	Value
Battery voltage (nominal)	43.2 V
Battery voltage (max)	50.4 V
Battery voltage (min)	30 V
ΔV_o	0.1 V
ΔV_i	0.5 V
$\Delta I_i n$	0.6 A
Efficiency	$>$ 90%

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Since the load is a battery, the output ripple voltage must be limited to avoid aging and heating. The industry standard is to limit the output voltage ripple to 0.5% of the nominal voltage. As for Therefore, the values presented in Table 4.2 were selected. Also, the voltage output voltage ripple must be low to avoid unexpected behavior of the electronic components that are also connected to the battery. As for the input (the solar panel) the voltage ripple must be low for easier measurement and precise control. As for the current, it must be low to reduce electromagnetic interference and losses.

4.1.2.A Inductor

The inductor is one of the most important components in the boost converter since it is responsible for storing energy and releasing it to the load.

The choice of an inductor will highly influence the performance of the converter. High inductance will generate low current ripple, however it generates losses due to its internal resistance. On the other hand a smaller inductance will generate high current ripples, high stress in components and high magnetic interference.

On the other hand, a smaller inductance will generate high current ripples that can cause the system to operate in Discontinuous Conduction Mode (DCM). In DCM the inductor current reaches zero during the switching period, which means that the energy stored in the inductor is not enough to supply the load during the off time. This can cause high current and voltage ripples, which can damage the components and reduce the efficiency of the converter. Therefore, it is important to ensure that the inductance is high enough to operate in Continuos Conduction Mode (CCM).

The equation that describes the inductor is defined as:

$$V_L = L \cdot \frac{dI_L}{dt} = L \cdot \frac{\Delta i_l}{\Delta t} \Leftrightarrow L = \frac{V_L \Delta t}{\Delta i_L} \tag{4.7}$$

In the first stage (transistor in conduction mode), the voltage in the inductor is equal to the input voltage and it occurs for $D \cdot T$ seconds. Therefore, the ripple current in the inductor can be defined as follows:

$$\Delta I_L = \frac{V_{in} \cdot D \cdot T}{L} \tag{4.8}$$

As for the current ripple, it can be described using the riple factor and the average current in the inductor as follows.

$$r = \frac{\Delta I_L}{I_{L,avg}} \tag{4.9}$$

$$I_{L,avg} = \frac{I_o}{1-D} \tag{4.10}$$

Using Equation 4.10, 4.9, 4.7 and 4.2, the following equation is achieved:

$$L = \frac{V_{in}(1-D)DT}{I_o \cdot r} = \frac{V_{out}(1-D)^2 DT}{I_o \cdot r} \tag{4.11}$$

The value of D that maximizes Equation 4.11 is 1/3, which results in the critical inductance formula:

$$MAX[(1-D)^2 \cdot D] = (1-1/3)^2 \cdot 1/3 = \frac{4}{27} \tag{4.12}$$

$$L_{critical} = \frac{4}{27} \cdot \frac{V_o}{I_o \cdot r \cdot f} \tag{4.13}$$

Using the maximum battery voltage (50.4 V) and the maximum current ($I_{pv} + Margin = 10\ A$), and a ripple factor of 2 which is equivalent to the CCM limit we achieve the following result:

$$L_{critical} = \frac{4}{27} \cdot \frac{50.4}{10 \cdot 2 \cdot 100000} = 3.73\mu H \tag{4.14}$$

This is a reference value, if it were used it would highly stress the components, generating heat, magnetic interference and high current ripples. Therefore, the calculation was repeated aiming the output current ripple of 0.6 A, which is the value defined in Table 4.2. Solving equation 4.9 in terms of the output current ripple, the following equation is achieved:

$$r = \frac{\Delta I_L}{I_{L,avg}} = \frac{\Delta I_{out} \cdot (1-D)}{\frac{I_{out}}{(1-D)}} = \frac{\Delta I_{out} \cdot (1-D)^2}{I_{out}} \tag{4.15}$$

$$L_{critical} = \frac{4}{27} \cdot \frac{50.4}{10 \cdot 0.4 \cdot 100000} = 18.66\mu H \tag{4.16}$$

The final value of the inductor was defines as 68uH 100 μH, which results in a maximum inductor ripple current of 1.26 A and an output current ripple of 0.53 A. refazer o calculo

To select the apropriate inductor, the rated current and the saturation current must be taken into account. The rated current is the maximum current that the inductor can handle without overheating. In this case the rated current must be higher that the maximum short circuit current of the biggest solar panel configuration (40 cells in series), which is 6.56 A. Some margin must be added to this value, so the rated current must be higher than 10 A.

The saturation current is the maximum current that the inductor can handle without saturating. When the inductor saturates, the inductance drops significantly, which can cause high current ripples and damage the components. Therefore, the saturation current must be:

$$I_{sat} > I_{in} + \frac{\Delta I_L}{2} = 6.56 + \frac{1.26}{2} = 7.19\ A \tag{4.17}$$

The inductor chosen was the 74437529203680 from Würth Elektronik, which has an inductance of 68 μH, a rated current of 16 A and a saturation current of 7.2 A ($|\Delta L/L| < 10\%$) and 10.6 A ($|\Delta L/L| < 30\%$). The saturation current is too close with the objective, however since the inductance is higher than what is needed, the inductance would still be acceptable even at -30% inductance.

Additionally, it has a low Equivalent Series Resistance (ESR) of only 10.3 $m\Omega$, high self resonant frequency of 5.7 MHz and high Operating voltage of 250 V.

4.1.2.B Capacitor

The output capacitor is responsible for smoothing the output voltage and reducing the voltage ripple. The capacitor electric characteristic can be described by the following equation:

$$i_c = C\frac{dv_o}{d_t} \tag{4.18}$$

Based on this equation, the capacitor value can be calculated based on the desired voltage ripple and the load current as follows:

$$i_c = C_o\frac{\Delta V_o}{\Delta t} = C_o\frac{\Delta V_o}{D \cdot T} \Leftrightarrow C_o = \frac{I_o D}{\Delta V_o \cdot f} = \frac{I_{in}(1-D)D}{\Delta V_o \cdot f} \tag{4.19}$$

Using the maximum input current of 10 A, a duty cycle of 0.5 (value that maximizes the formula), a voltage ripple of 0.1 V and a switching frequency of 100 kHz, the following value is achieved:

$$C_o = \frac{10 \cdot (1-0.5) \cdot 0.5}{0.1 \cdot 100000} = 250\mu F \tag{4.20}$$

The same can be done for the input capacitor, but in this case the voltage ripple is not as critical as in the output capacitor, since the input voltage is not directly connected to the load. Therefore, a voltage ripple of 0.5 V was used:

$$C_{in} = i_{in} \cdot \frac{\Delta t}{\Delta V_i} = \frac{i_{in} \cdot D}{\Delta V_{in} \cdot f} = \frac{10 \cdot 0.5}{0.5 \cdot 100000} = 100\mu F \tag{4.21}$$

To choose the appropriate capacitors, the voltage rating, current ripple and ESR must be taken into account. The voltage rating must be higher than:

$$V_{rating} > MAX(V_{in}) + \Delta V_{in} = 28.5 + 0.5 = 29\ V \tag{4.22}$$

Where 28.5 is the maximum voltage of a solar panel (40 cells in series) and 0.5 is the voltage ripple defined in Table 4.2.

The current ripple must higher than the RMS input current, which can be calculated based on the RMS formula for triangle wave as follows:

$$I_{ripple} > \frac{\Delta I/2}{\sqrt{3}} = \frac{1.26/2}{\sqrt{3}} = 0.36\ A \tag{4.23}$$

As for the ESR the value must be as low as possible to reduce the power losses.

After analyzing the market, the input capacitor selected was the 870575875006 which is a 56 μF (2 in parallel) with a voltage rating of 63 V and a ripple current of 2.4 A each (total of 4.8 A). The main reason for this choice was his low ESR (22 $m\Omega$) compared to other capacitors with similar values. Additionally, by using two capacitors in parallel, the ESR is reduced even further and the current ripple that they with stand is higher and the reliability of the system is increased.

The same process was repeated for the output capacitor, but in this case the voltage rating must be higher than the maximum battery voltage, which is 50.4 V. The current ripple must be higher than the RMS output current, which can be calculated as follows:

$$V_{rating} > MAX(V_{out}) + \Delta V_{out} = 50.4 + 0.1 = 50.5\ V \tag{4.24}$$

$$I_{ripple} > \sqrt{\frac{I_{in}^2}{4} + \frac{\Delta I^2}{12} \cdot (1 - D)^2} = \sqrt{\frac{6.56^2}{4} + \frac{1.28^2}{12} \cdot (1 - 0.27)^2} = 3.29\ A \tag{4.25}$$

The ripple current in the output capacitor is much higher than in the input capacitor, and it is not a common value for electrolytic capacitor, therefore instead of using a single capacitor, multiple capacitors in parallel will be used to achieve the desired current ripple rating.

After simulations and analyzing the market, the output capacitor value were increased due to the output current ripple dependency on the output voltage ripple. The final capacitor chosen was the 860040875005 with 100 μF (5 in parallel, so 5x100 μF) with a voltage rating of 100V and a ripple current of 0.8 A each (total of 4 A). Some other options were considered, and compared for optimization in section 4.1.3.B.

Additionally, the output and input needs to filtered to reduce the high frequency noise generated by the switching. This is achieved by adding a small capacitor in parallel with the main capacitors. The values use are 1 μF and 0.1 μF. For this use case ceramic capacitors are better since their stacked multilayer construction provides extremely low ESR and Equivalent Series Inductance (ESL) which over all let them maintain low impedance up into the MHz range.

4.1.2.C Transistor and gate driver

The transistor is the switching device on a synchronous boost converter. The choice of the transistor will highly influence the behavior and performance of the converter.

When selecting a transistor for a switching converter, not only the maximum rating mater but the switching speed and internal resistance are also quite important.

There are many types of transistors (IGBT, MOSFET, GaN, etc.), however, at this frequency (100 kHz) and power level the choice is between MOSFETs and GaN transistors (Figure 4.3). The GaN (Gallium Nitride) transistor technologies has appeared in 1993 in lab experiments, however it has only reached the market more recently, first for radio frequency products in 2004 and only in 2009 for power electronics. The advantage of GaN transistors is their low internal resistance and very fast switching speed, which can significantly reduce the switching losses and therefore increasing the efficiency of the converter.

Additionally, they have a high energy density and lower Vgs(on). However, they are more expensive and require a precise gate driver to operate correctly because there max(Vgs) is usually low.

In applications where the efficiency is a key factor in the design like solar inverters or Maximum Power Point Tracker (MPPT), it is better to use the component that has lower Internal (On) Resistance (RdsOn) and supports higher switching frequency. In this case, GaN would be a better fit. MOSFET or Silicon Carbide MOSFET would take the second place since MOSFETs can handle higher switching frequencies as compared to IGBTs [9].

There are a lot of options in the market, however there was one in specifically that standout due to their compact size, while having integrated gate driver and two transistors in the same package. The Integrated Circuit (IC) is the EPC23102, which was designed for power applications, with an incredibly low RdsOn (only 5.2 $m\Omega$), 35 A of maximum power and a maximum operating voltage of 100 V.

Additionally, it has incorporates shoot-through protection that protects the transistor from a control error, however a dead time needs to be introduced in software. It uses a DFN package which is good for heat dissipation and can always be attached to a heat sink for better performance.

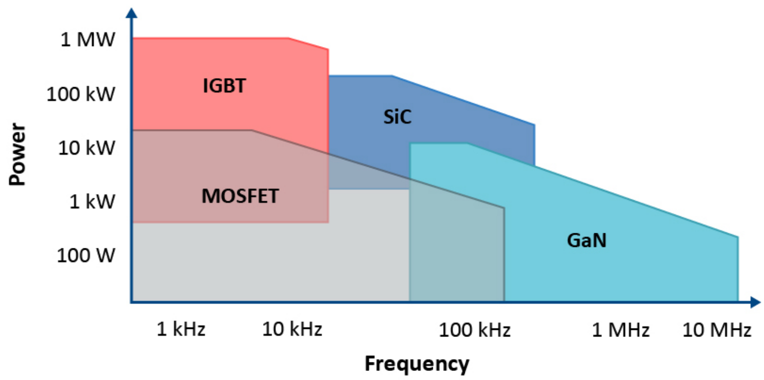

Figure 4.3: Technologies selection based on power and switching frequency [9].

4.1.3 Simulations

Objective

4.1.3.A Solar Panel model

The simulations started at LTspice with the aim to test the converter at different conditions. To achieve that, a Photovoltaic (PV) electrical model of the cells were designed using a five parameter model (1M5P) configuration. There are more electrical models that can simulate a solar panel, some with more accuracy but more complex. This model was perfect for the aim of the simulation since it is both reliable and simple.

The 1M5P model represents the solar cell through a single diode equivalent circuit composed of a current source in parallel with one diode, a series resistance R_s, and a shunt (parallel) resistance R_{sh}, as shown in Figure 4.4. The current source represents the photogenerated current I_{ph}, which is proportional to the incident irradiance, while the diode accounts for the non linear I-V characteristic of the p-n junction. The series resistance models the losses associated with the material bulk resistance, metal contacts and interconnections, whereas the shunt resistance represents the leakage currents through the cell, typically caused by manufacturing defects.

The output current of the model is given by the characteristic equation 4.26, where I_0 is the diode saturation current, n is the diode ideality factor, and V_t is the thermal voltage [31]. Together with I_{ph}, R_s and R_{sh}, these constitute the five parameters that give the model its name.

$$I = I_{ph} - I_0 \left[\exp\left(\frac{V + IR_s}{nV_t} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \tag{4.26}$$

These parameters are not directly provided by manufacturers, but can be extracted from the standard datasheet values, namely the open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), voltage and current at the Maximum Power Point (MPP) (V_{mp}, I_{mp}), and the respective temperature coefficients, through a set of equations that ensure the model matches these reference points under Standard Test Conditions (STC). There are several ways to do it, however it is outside the range of this work. To simplify, calculations were carried using online tools and will not be explained in detailed here. se achar necessario posso escrever as equações aqui

This approach allows the panel behavior to be accurately reproduced under different irradiance and temperature conditions, while keeping the circuit implementation simple enough to be integrated directly into LTspice.

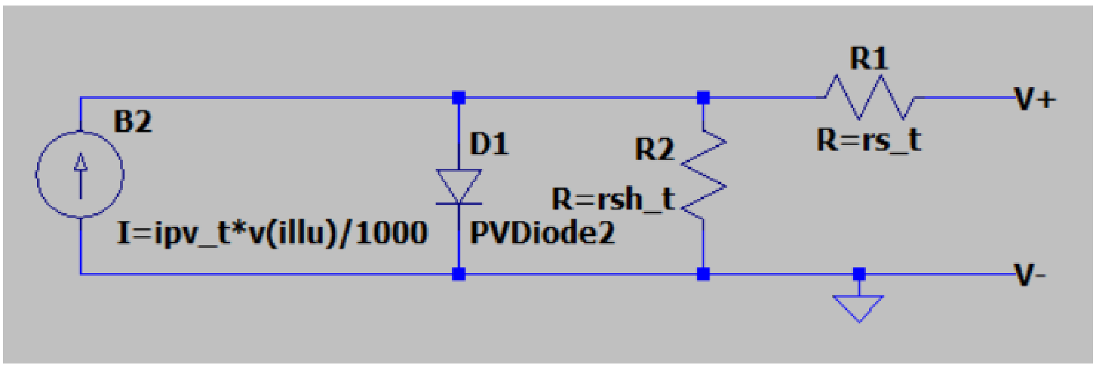

Figure 4.4: Solar panel model diagram (1M5P configuration).

4.1.3.B Boost Converter

To validate the theoretical calculations, simulations in LTspice were carried using the boost converter in steady state mode and a fix duty cycle.

Considering a solar panel with 35 cells, five points of CC-CV charging curve were chosen to test the performance of the converter. These five points are represented in Figure 4.5.

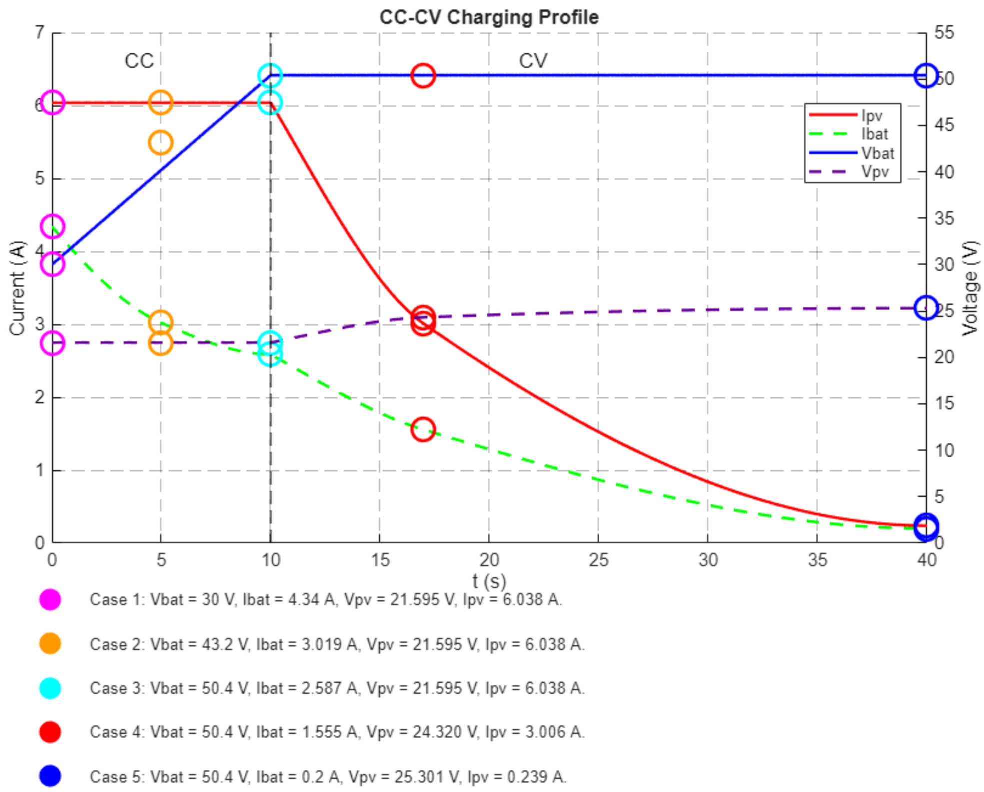

Figure 4.5: CC-CV charge characteristics and analysis point selection.

It is important to note that in this CC-CV curve the constant current is given by the solar panel and not by the battery as it is usually expected. This is only truth due to the maximum charge current of the battery being lower than the current that can be provided by the solar panel. So at CC, the converter is working at MPP.

With this five points chosen, some simulations were carried using capacitors and inductors that would fit the requirements calculated earlier. The objective of these simulations is to optimize the performance of the converter, leading to better choices.

The model was built in LTspice with approximated models given by the manufactures for higher precision. Only the solar panel model was designed by hand, as previously mentioned.

The battery was simulated using a voltage source and a resistor representing the internal resistance of the battery.

The dead time of the transistors was implemented too. Using laboratory measurements, the rising and falling timing was measured and used in simulation. Has for the dead time, it is used the same as in the final prototype.

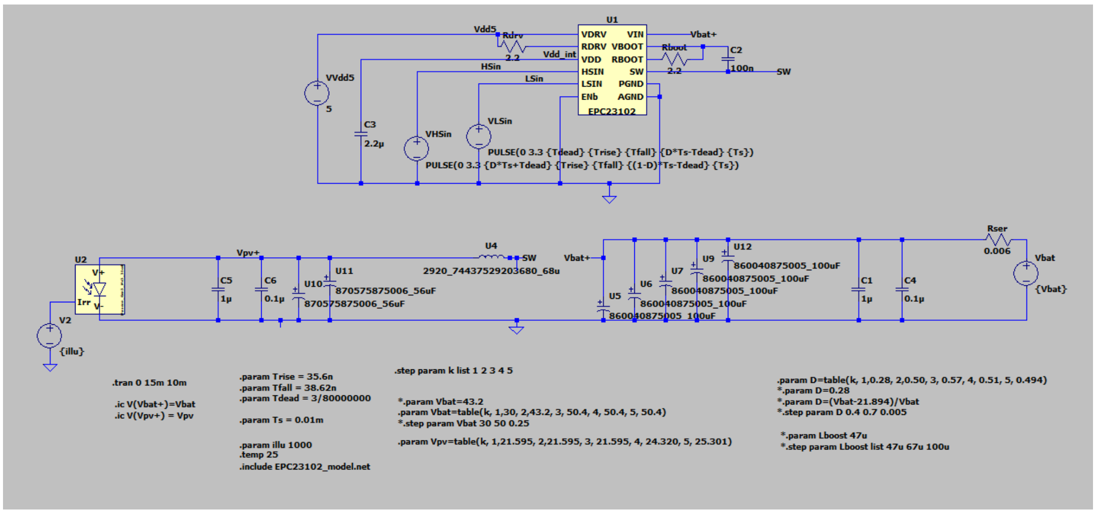

Figure 4.6: Simulation circuit schematic.

The first simulations were carried to choose the input and output capacitors. The idea is to simulate the circuit using a known inductor (68 uH) and change the capacitors for different configurations. All configurations are overdimensioned to give low ripples (current and voltage). So the main factor is the efficiency of the converter.

In Table 4.3 is shown the results. The efficiency of all combinations are very high however the highest is the one with two 56 uF as input capacitors and tree 220 uF capacitors as output capacitors. Unfortunately the 220 uF were not available at the moment, so the second best option was chosen, the five 100 uF capacitors.

It is worth noting that this is a simulation and therefore the efficiency is higher than what will be measured in laboratory results. Additionally, the difference between each combination it is very small. To make the product cheaper the third option would be as great as the one chosen.

Table 4.3: Loss (%) for different capacitor configurations (Cin, Cout).

Case #	2x56 μF, 3x220 μF	100 μF, 3x220 μF	100 μF, 5x100 μF	2x56 μF, 5x100 μF
1	0.5729	0.5796	0.5856	0.5788
2	0.6762	0.6985	0.7069	0.6844
3	0.7359	0.7655	0.7743	0.7449
4	0.6654	0.7091	0.7161	0.6735
5	3.4177	3.7148	3.7554	3.4671

Similar to the previous simulation, the inductor was chosen by keeping all components with the same values (the capacitors used were Cin = 2x56uF and Cout = 5x100uF) and changing the inductor. Similar to before, all inductors simulated are within the minimum required.

The losses for each case are represented in Table 4.4. Similar to before, the efficiency difference is very low but not as linear as in the previous simulation, likely due to the inductor models chosen.

The inductor chosen will highly influence the input ripple which can generate heat, and electrical interference. Table 4.5 shows the different ripple currents for each inductor in each case of the CC-CV curve. The 22 uF inductor showed higher ripple current reaching values close to the input current. The inductor chosen was the 68 uF since it has overall good efficiency and reasonable input current ripple.

Table 4.4: Loss (%) for different inductance values.

Case #	22 μH	33 μH	47 μH	68 μH	100 μH
1	0.4206	0.4439	0.5453	0.5788	1.2659
2	0.6645	0.6793	0.6627	0.6844	1.3664
3	0.7883	0.8076	0.7290	0.7449	1.4193
4	1.0556	1.0583	0.6867	0.6735	0.9624
5	6.7045	6.6230	3.6556	3.4671	3.8177

Table 4.5: Δi_{in} $(I_{in,\max} - I_{in,\min})$ for different inductance values.

Case #	22 μH	33 μH	47 μH	68 μH	100 μH
1	2.6939	1.8906	1.3256	0.9781	0.6251
2	4.7934	3.3416	2.3354	1.7066	1.0707
3	5.4733	3.8253	2.6813	1.9644	1.2839
4	5.5596	3.8565	2.7075	2.0139	1.2925
5	5.5520	3.8510	2.6903	1.9636	1.2344

Now that the converter components are all chosen, the following measurements were taken in LT-spice:

Table 4.6: Ripple and Power Loss Summary for Each Operating Step

Case #	ΔV_{bat} (V)	ΔV_{pv} (V)	ΔI_{in} (A)	Loss (%)
1	0.0406	0.0145	0.9781	0.578
2	0.0491	0.0212	1.7066	0.684
3	0.0529	0.0244	1.9644	0.745
4	0.0406	0.0220	2.0139	0.674
5	0.0302	0.0215	1.9636	3.464

Iin é difrente de IL. O Iin tem quase nada de ripple

Both voltage ripples are lower than what was defined as the goal in Table 4.2. As for the current ripple it is still reasonable. The losses/efficiency are also higher than what was aimed, reaching values up to 99.4%, in simulation.

A voltage sweep were performed, and as expected the efficiency drop with the increase of output voltage (Figure 4.7). Also, as mentioned earlier, the output current drop because in Constant Current

(CC) mode the solar panel current is constant, so the output current drops with the increase of battery voltage due to the power conservation principle.

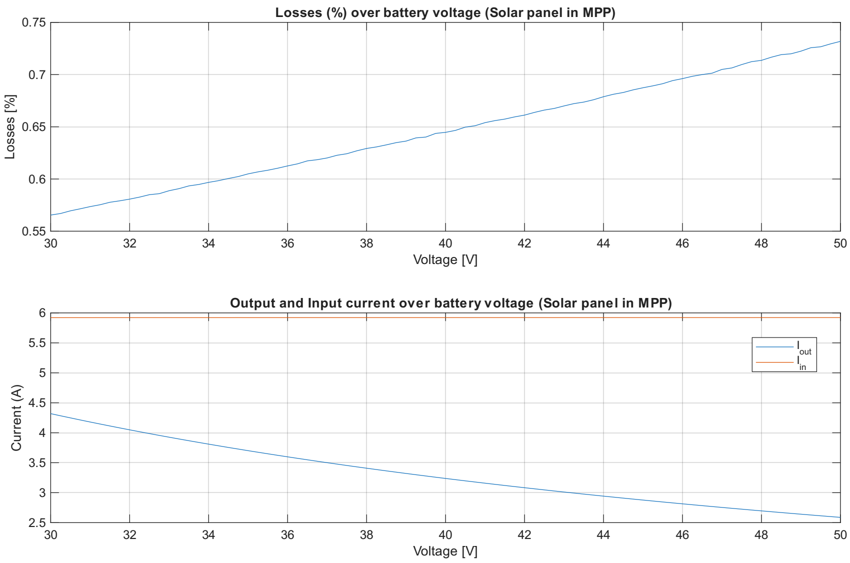

Figure 4.7: Battery voltage sweep simulation results.

Note: in Section 4.6.1 it is mention the addition of two shotcky diodes that were not considered in simulations. These diodes slightly increase the efficiency of the converter (+0.03%) due to their lower forward voltage.

Add wave forms??!?!?!

4.1.4 Power Loss Estimation

Although LTspice provides an accurate estimation of the converter efficiency, it does not directly identify the contribution of each component to the total power dissipation. Therefore, an analytical power loss model was developed in MATLAB to estimate the individual losses of each component of the proposed synchronous boost converter. The model uses the electrical parameters of the selected components obtained from their datasheets together with the converter operating conditions.

The total converter losses are obtained by summing the losses associated with the GaN transistors, inductor, capacitors, dead time and gate driver,

$$P_{loss} = P_{cond_{HS}} + P_{cond_{LS}} + P_{sw_{HS}} + P_{sw_{LS}} + P_L + P_C + P_{dead} + P_{gate}. \tag{4.27}$$

The converter efficiency is then calculated as

$$\eta = \frac{P_{out}}{P_{out} + P_{loss}} \times 100\%. \tag{4.28}$$

The conduction losses of each GaN transistor are given by

$$P_{cond} = I_{RMS}^2 R_{DS(on)}. \tag{4.29}$$

These losses are primarily caused by the on state resistance of the transistors, resulting in power dissipation whenever current flows through the channel.

The switching losses are estimated using Equation 4.30, which takes into account the loses during the period when the transistor is not fully nor fully off. Also, the losses associated with charging and discharging the output capacitance of each transistor.

$$P_{sw} = \frac{1}{2} V_{out} I_{L_{AVG}} (t_r + t_f) f_{sw} + \frac{1}{2} C_{oss} V_{out}^2 f_{sw} \tag{4.30}$$

The dead-time losses are calculated using Equation 4.31. These losses originate from the conduction of the reverse conduction path during the dead-time interval required to prevent shoot through. This reverse condition path is either from the third quadrant conduction from the GaN device or from the parallel schottky diode added later in version 1.1 (this is explained in Section ?)

$$P_{dead} = 2 \cdot V_f \cdot I_{L_{AVG}} \cdot t_{dead} \cdot f_{sw}. \tag{4.31}$$

The inductor and capacitor losses are estimated using the RMS current passing through them and their ESR

$$P_L = I_{L_{RMS}}^2 ESR. \tag{4.32}$$

$$P_C = I_{C_{RMS}}^2 ESR, \tag{4.33}$$

Although magnetic core losses of the inductor could be estimated too, the selected manufacturer does not provide the required magnetic material parameters. Consequently, only the winding copper losses are considered, leading to a slight underestimation of the total converter losses.

Finally, the gate driver losses are estimated using Equation 4.34 which takes into account the energy needed to drive the transistors. Although the EPC23102 has an internal gate driver, this is still relevant.

$$P_{gate} = (Qg_{LS} + Qg_{HS}) * Vdrv * fsw. \tag{4.34}$$

Figure 4.8 presents the resulting power loss distribution. The calculated total power loss is approximately 1.188 W, corresponding to an efficiency of 99.10%, which agrees well with the LTspice simulation results presented in the previous section.

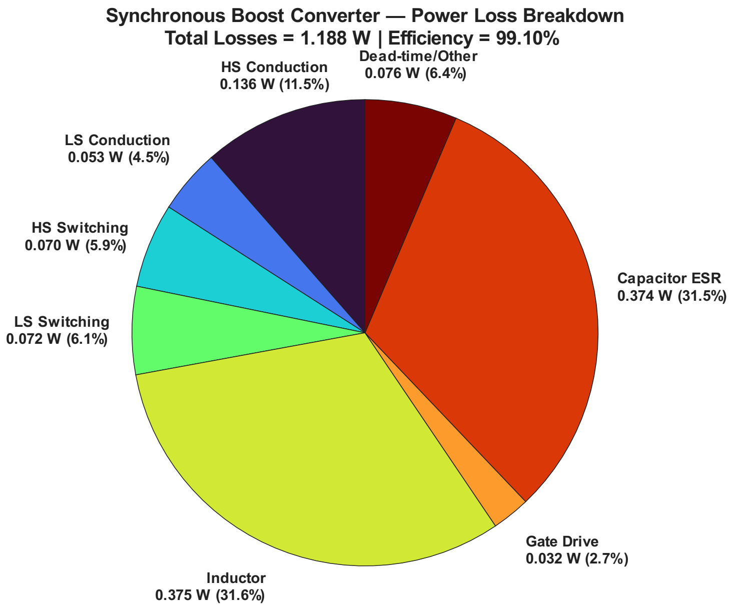

Figure 4.8: Analytical power loss distribution of the synchronous boost converter for Case 1.

The largest contributors to the total losses are the inductor copper losses and the capacitor ESR losses, each accounting for approximately 31.5% of the total power dissipation. This highlights the importance of selecting an inductor with low winding resistance and capacitors with low ESR, as these passive components dominate the converter losses.

The conduction losses of the GaN transistors account for approximately 16% of the total losses, showing the importance of choosing transistors with low conduction resistance. The high-side device dissipating more power than the low-side transistor due to its longer conduction interval. Switching losses represent approximately 12% of the total losses, including both transition losses and the energy required to charge and discharge the output capacitances. Dead-time losses contribute approximately 7%, while the gate driver losses are almost negligible (only 2.7%).

Additionally, all Printed Circuit Board (PCB) connections have an inductance that will represent losses too.

Overall, the analysis indicates that nearly two-thirds of the converter losses originate from the passive components, whereas the semiconductor devices account for approximately one-third. It should be noted that the presented values are based on analytical equations and datasheet parameters. Since magnetic core losses were not considered, the calculated efficiency represents a slight overestimation of the actual converter performance.

However, both simulation and theoretical analysis indicate that the choice of each components lead to a highly efficient converter.

4.2 Processing unit

Maybe i will remove this part { Several types of processing units can be considered for this project, including Microcontroller Units (MCUs), Field Programmable Gate Arrays (FPGAs), and Application-Specific Integrated Circuits (ASICs).

ASICs are custom-designed integrated circuits optimized for specific applications, offering high efficiency and performance. However, they are typically expensive and time-consuming to design and manufacture. Although commercially available ASICs, commonly referred to as ICs, mitigate these drawbacks, they are usually made for a specific application and therefore not suitable for the high performance and flexibility required in this project, as discussed in Section 3.6.1.

FPGAs are reconfigurable devices capable of implementing custom digital circuits through programmable logic blocks based on look-up tables (LUTs) and interconnections [32]. Their main advantage lies in their inherent parallel processing capability, which enables high-performance operation. Nevertheless, FPGAs generally present high costs, high design complexity, and volatile configuration memory, requiring reprogramming after power loss [33, 34].

In contrast, MCUs are widely adopted in embedded systems due to their low cost, ease of use, and versatility. They integrate processing, memory, and peripherals in a single non-volatile device, simplifying hardware design and development. Although their sequential execution limits parallel processing performance compared to FPGAs, modern MCUs provide sufficient computational capability for many control and signal-processing tasks [33].

Considering the emphasis on low cost, reduced development time, and implementation simplicity, a MCU is selected for this application. Furthermore, the performance of current MCUs is adequate to meet the requirements of the MPPT algorithm and the battery charging strategy.

——————}

A Microcontroller Unit (MCU) typically consists of three fundamental components: a central processing unit (CPU), memory, and peripheral modules. The CPU executes program instructions and performs arithmetic and logical operations. Memory is generally divided into non-volatile memory (Flash) for program storage and volatile memory (SRAM) for data and runtime variables. Peripheral modules provide interfaces to the external environment and may include analog-to-digital converters (ADCs), timers, pulse-width modulation (PWM) units, communication interfaces (such as UART, SPI, I²C and CAN), and general-purpose input/output (GPIO) ports. The integration of these components enables standalone operation without the need for extensive external circuitry [35].

When selecting a MCU for a specific application, the first factor to consider is to use an 8, 16, or 32 bit MCU. For this application, a 16-bit MCU would probably be sufficient, however to ensure future-proof and compatibility with other Técnico Solar Boat (TSB) systems, the choice is for a 32-bit MCU.

Besides, nowadays, 32-bit MCUs are considered the industry standard since most embedded system Research and Development (R&D) effort is focused on 32-bit cores, and thus both architectures can achieve similar power consumption [36]. There are two main architectures for 32-bit MCUs: AVR and ARM. Being ARM, the most widely used architecture in the industry and the most advanced, it was chosen for this project [33].

STM32 family of MCUs offers a wide range of solutions with different characteristics, making it suitable for various applications, including this project. This family of microcontrolers is used by TSB, so it will be used for this project to maintain the standard.

The MCU requirements for this project are:

- Fast clock frequency: 80 MHz or higher;
- Support 1x CAN peripheral, 1x I2C, 1x UART;
- Able to read 4 or more analog signals;
- 12bit ADC or more.

The final decision was the STM32L476. Although it is not the most suitable STM32 microcontroller for this application, it was selected due to its low cost and because it was already widely used within TSB, reducing development time and associated risks.

The STM32L476 is based on an ARM Cortex-M4 core with floating-point support and operates at frequencies up to 80 MHz. The device provides 1 MB of Flash memory and 128 kB of SRAM, which is more than sufficient for the implementation of the MPPT algorithm, data acquisition, communication protocols, and debugging features. Regarding peripherals, it includes three 12-bit ADCs with up to 16 external channels each, multiple general-purpose and advanced timers capable of high-resolution PWM generation, several communication interfaces (CAN, SPI, I^{2}C, UART, USB), and DMA controllers that allow peripheral operation with minimal CPU intervention.

These features make the STM32L476 particularly attractive for power electronics applications, where simultaneous voltage and current measurements, fast PWM generation, and real-time control execution are required. Furthermore, the presence of a hardware floating-point unit simplifies the implementation of control algorithms and numerical calculations.

If there were no limitations, a microcontroller with a higher clock frequency would have been selected, increasing both the PWM duty-cycle resolution and the available computational power. Additionally, a device with native 16-bit ADCs would improve measurement accuracy and reduce quantization effects. Nevertheless, the STM32L476 provides sufficient performance and peripheral resources to meet the requirements of this application.

4.3 Sensors

4.3.1 Voltage and current sensing

Accurate voltage and current measurements are essential for MPPT operation, since tracking algorithms directly rely on these quantities to estimate power and adjust the operating point of the PV array. Measurement noise or inaccuracies directly impact tracking efficiency, especially in high frequency switching DC-DC converters.

Voltage sensing is generally straightforward and can be implemented using resistive voltage dividers followed by ADC. Additionally, it is suggested to use an op-amp as a buffer to reduce the impact of the ADC input capacitor, while charging. This buffer reduces noise and improves the accuracy of the measurement while also allowing the use of higher sampling rates.

Current sensing, however, is more critical and requires careful consideration of the available measurement techniques.

The two main techniques are:

- Hall effect current sensor.
- Shunt resistor sensing with operational amplifiers;

The hall effect current sensor is a non-invasive method that uses the Hall effect to measure current. It provides galvanic isolation and can do bidirectional measurements. However, it is typically more expensive, has lower bandwidth, and may introduce additional noise due to its magnetic sensing nature.

As for the shunt resistor sensing, it is a simple and cost-effective method that uses a low-value resistor to convert the current into a voltage drop. This voltage is then amplified using operational amplifiers to match the ADC input range. This method provides high accuracy, low cost, and fast response time, but it requires careful design to minimize power loss since it will dissipate power proportional to the current.

In the end product a IC based on a shunt resistor sensing method was used, which as a built-in amplifier. The sensor chosen was the INA253A2IPWR, which has a PWM rejection block and small form factor that will be extremely useful for this application. However, this IC is more expensive than a simple shunt resistor and op-amp solution, but it is more accurate and has a better noise rejection. In the future it can be swapped for a lower cost solution, if the performance is acceptable.

Both sensors use the internal ADC from the STM32L476, which has a 12 bit ADC (with 3.0V as the reference). The current sensor output an analog voltage that is equal to the current multiplied by 0.2 V/A. Which means that the resolution of the current measured is 3.0/(4096 * 0.2) = 3.66 mA. As for the voltage sensor, it outputs an analog voltage that is equal to the voltage multiplied by 10/210. The resolution is 21 * 3.0/(4096) = 15.38 mV, which is more than enough for this application.

Both circuits are represented in Appendix 4.6. place the apendix reference

4.3.2 Temperature and Irradiance sensors

For security reasons it is important to monitor the temperature of the PCB. Also, the temperature and irradiance of the PV panel are important to monitor for performance analysis and defect detection. Also, this measurements can be used to improve the MPPT algorithm, since the maximum power point is dependent on temperature and irradiance.

For temperature sensisng there is a wide range of options, from simple thermistors to more complex digital temperature sensors. For this application a simple thermistor was used, since it is cheap and easy to implement. The thermistor is connected to a voltage divider and the voltage is read by the ADC. The thermistor used to measure ambient temperature and transistor temperature was the ERT-J1VS104FA, which has a resistance of 100kΩ at 25℃ and a B value of 4330K. As for the thermistor used to measure the PV panel temperature, it was the 104JT-025, which has a resistance of 100kΩ at 25℃ and a B value of 4390K. The temperature can be calculated using the Steinhart-Hart equation [37].

$$\frac{1}{T} = A + B \cdot ln(R) + C \cdot ln(R)^3 \tag{4.35}$$

This equation uses 3 parameters (A, B and C) that are usaly not provided in data sheet but can be calculated using 3 measurements of temperature and resistance. However, the most common approach is to use a simplified version of the equation, which uses only 2 parameters (B and R0). The temperature can be calculated using the following equation:

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{B} \cdot ln(\frac{R}{R_0}) \tag{4.36}$$

As for the irradiance sensor, the TSL2591 was used, which is a digital light sensor with a wide dynamic range and high sensitivity. It can measure light intensity from 0.000118 lux to 88,000 lux, making it suitable for various lighting conditions. The sensor communicates with the MCU via the I2C protocol, allowing for easy integration and data acquisition.

Since it measures the light intensity in lux, it is not a direct measurement of irradiance and it needs to be calibrated to convert the lux value to irradiance in W/m$\hat{2}$. This calibration was done using a irradiance sensor from ??? which measures the irradiance in W/m$\hat{2}$. The calibration was done by measuring the lux and irradiance at the same time and calculating the conversion factor. The conversion factor was found to be ???? W/m$\hat{2}$ per lux at laboratory conditions. This factor changes with the spectral distribution of the light source, so it would need be calibrated for different light sources.

There are bether options for irradiance sensors, but this sensor was chosen due to its low cost and easy integration with the MCU. The main disadvantage of this sensor is that it is not a direct measurement of irradiance and it needs to be calibrated. However, for laboratory testes it is sufficient since the objective is to record when does the irradiance change to analyze the performance of the MPPT algorithm. For precise measurements of irradiance, the sensor used to calibrate the TSL2591 can be used, which is a more expensive sensor that provides direct measurements of irradiance but does not provide communication protocol to record the data.

In the future, if the irradiance value is needed to be used in the MPPT algorithm, a more precise irradiance sensor is needed. The most precise sensors use a PV cell, which based on his temperature, short circuit current and open circuit voltage can provide a good estimation of the irradiance. My suggestion is to design this circuit and use the same cell that is used in the PV panel to measure the irradiance. This way, the irradiance measurement will be accurate and with exactly the same spectral distribution as the PV panel which will increase the performance of the MPPT algorithm. This was not implemented in this project due to time constraints, but it is a good option for future work.

4.4 Software

4.4.1 MPPT algorithm

simulação em Matlab..

4.4.2 Firmware

Tool de STM32 usada, flow chart

Flag-based scheduler, Timers, PWM resolution, ADC mode, APP conection, CAN com, serial comm, watchdog,...

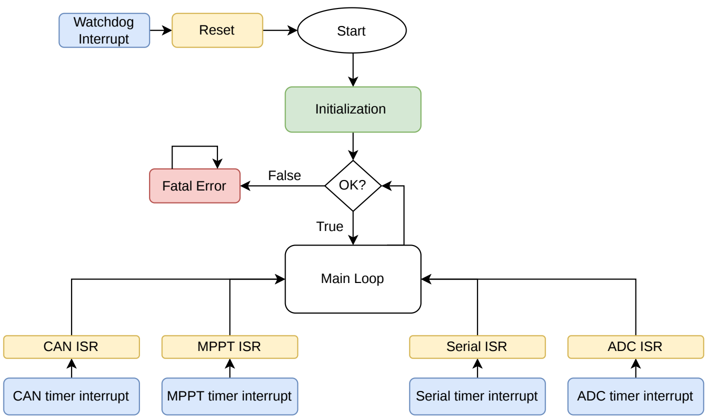

Figure 4.9: .

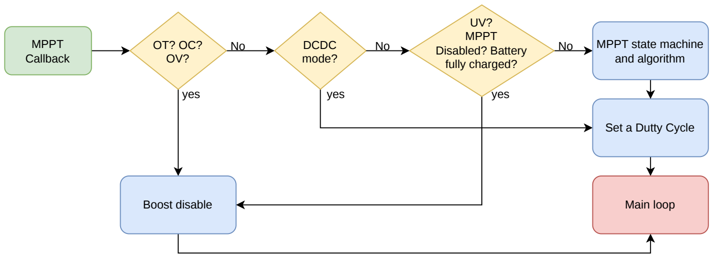

Figure 4.10: .

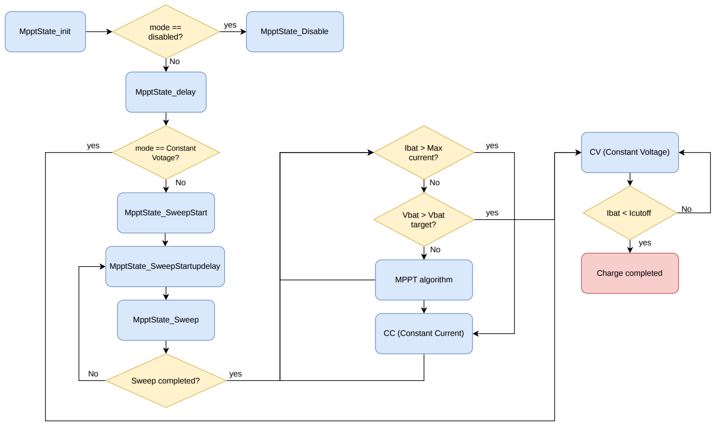

Figure 4.11: .

4.4.3 Graphical User Interface

Funcionalidade, protocoolo de comunicação, como foi feita, Qt widget, etc.

4.5 Precharge Circuit

The precharge circuit is used to safely energize high-capacitance loads by limiting inrush currents during system startup. This section introduces the problem of inrush currents, discusses the adopted mitigation strategy, and presents the final circuit implementation.

4.5.1 Problem

In the automotive sector, high-capacitance loads such as inverters, motor controllers, and DC/DC converters are commonly encountered. When these systems are connected to a power source, their large input capacitance can lead to significant inrush currents during the transient period. These currents, which can reach the order of kiloamperes, may generate electrical arcing, stress or damage the contactors, and in severe cases cause contact welding, preventing proper disconnection of the circuit. Additionally, such high transient currents can also stress or damage the load itself, particularly sensitive power electronics at the input stage, potentially leading to component failure or reduced operational lifetime.

In the context of the MPPT, the battery is connected and disconnected through a relay depending on the system operating conditions, such as the battery state of charge. When the relay closes, the input capacitors of the MPPT act as an initially uncharged capacitive load, resulting in a high transient current as they are charged from the battery.

As an example, consider the battery operating at its maximum voltage ($V_{bat} = 50.4\ V$) connected to a single MPPT with an input capacitance of 500 μF. Initially, the capacitors are fully discharged ($V_c = 0\ V$). The total series resistance of the circuit is composed of the internal battery resistance ($R_{bat} = 6\ m\Omega$), the equivalent capacitor resistance, and the resistance of the connection cables.

The cable resistance can vary depending on length and cross-section, but it is typically small and is therefore neglected in this analysis. The equivalent resistance of the capacitors is given by the resistance of a single capacitor ($R_{cap} = 200\ m\Omega$) divided by the number of capacitors in parallel ($n = 5$), resulting in:

$$R_{caps} = \frac{200\ m\Omega}{5} = 40\ m\Omega \tag{4.37}$$

The total circuit resistance is therefore:

$$R_{total} = R_{bat} + R_{caps} + R_{cables} \approx 46\ m\Omega \tag{4.38}$$

The current flowing into the capacitors at any time can be described by:

$$i(t) = \frac{V_s - V_c(t)}{R} \tag{4.39}$$

From this, the worst-case inrush current occurs at $t = 0$, when $V_c = 0$:

$$i(0) = \frac{50.4}{0.046} = 1097\ A \tag{4.40}$$

Such current levels can severely degrade or destroy the contactors over time. Even if immediate failure does not occur, the repeated exposure significantly reduces their operational lifetime.

This issue becomes even more critical when multiple MPPTs are connected in parallel. In the São Gabriel 01 (SG01) system, four MPPTs are used, resulting in a total capacitance of 2000 μF and a reduced equivalent resistance of approximately 16 mΩ. In this case, the inrush current increases to approximately 3150 A, further emphasizing the necessity of a precharge strategy.

4.5.2 Solution

A common approach to mitigate inrush current in capacitive loads is to introduce a controlled precharge phase, in which the load is temporarily connected to the power source through a series resistance. This resistance limits the initial charging current, allowing the capacitors to charge gradually. Once the voltage across the load reaches a predefined threshold, the resistor is bypassed and the system transitions to normal operation, eliminating the associated power losses.

Figure 4.12 illustrates the fundamental concept of the proposed precharge circuit. In this topology, K3 is used to connect the load through a precharge resistor, while K2 is responsible for bypassing the resistor after the precharge phase is completed. The optional contactor K1 may be used to provide galvanic isolation between the battery and the load when required.

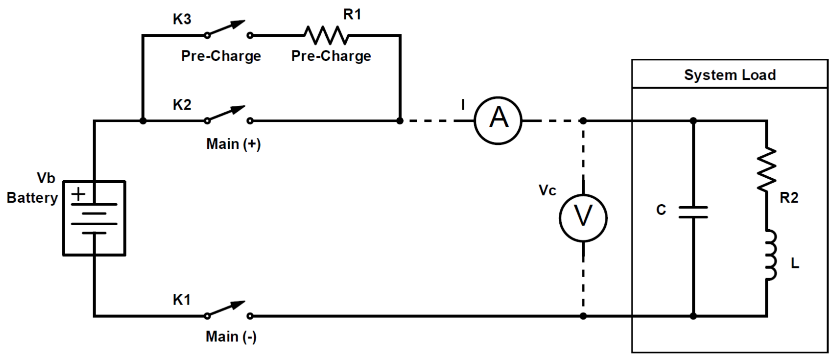

Figure 4.12: Typical precharge circuit [10].

This strategy ensures that the initial charging current is limited by design, significantly reducing electrical stress on the contactors and load, therefore improving system reliability. Once the precharge phase is completed, the resistor is removed from the current path, ensuring that no additional conduction losses are introduced during normal operation.

4.5.3 Design

The precharge process can be modelled as an RC charging circuit, where the load capacitance is gradually charged through a series resistance. The voltage across the capacitor during the precharge phase is given by:

$$V_c(t) = V_s\left(1 - e^{-t/RC}\right) \tag{4.41}$$

Combining Equation 4.41 with Equation 4.39, the instantaneous current during the precharge process can be expressed as:

$$i(t) = \frac{V_s e^{-t/RC}}{R} \tag{4.42}$$

These expressions show that both the current and voltage evolution are governed by the time constant $\tau = RC$, which directly determines the precharge duration and the peak inrush current. After one time constant ($t = \tau$), the current decreases to approximately 36.7% of its initial value, while the capacitor voltage reaches approximately 63.2% of the final value, as illustrated in Figure 4.13.

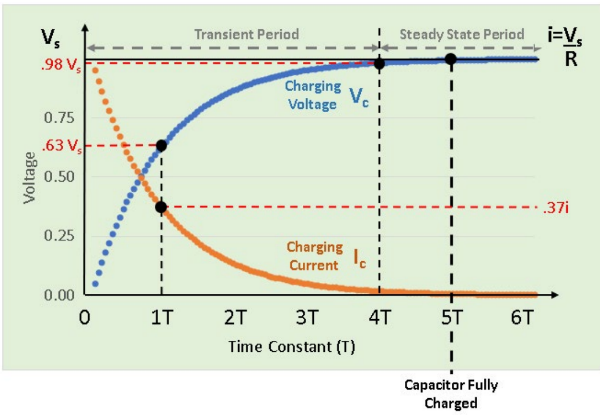

Figure 4.13: Precharge current and voltage evolution [10].

The transient period is typically considered to span from 0τ to 4τ, during which the capacitor reaches approximately 98.17% of its final voltage. Beyond this point, the system enters the steady-state region, and at approximately 5τ, the capacitor is effectively fully charged (99.33%), although the charging process is asymptotic and never reaches exactly 100%.

In practical implementations, the switching between precharge and normal operation is typically performed after approximately 4τ to minimise stress on the contactors while ensuring a sufficiently high load voltage.

Based on this criterion, the resulting inrush currents can be estimated as:

$$I_{inrush}(1 \times MPPT) = \frac{50.4 - 50.4 \cdot 0.9817}{0.046} = 20.05\ A \tag{4.43}$$

$$I_{inrush}(4 \times MPPTs) = \frac{50.4 - 50.4 \cdot 0.9817}{0.016} = 57.65\ A \tag{4.44}$$

To achieve the desired precharge threshold, a differential comparator circuit was implemented using two voltage dividers as inputs. These dividers scale both the battery-side and load-side voltages to a range compatible with the comparator while simultaneously defining the switching threshold corresponding to the target precharge percentage. The implemented circuit is shown in Figure 4.14. The precharge threshold can be adjusted by modifying the value of resistor R2.

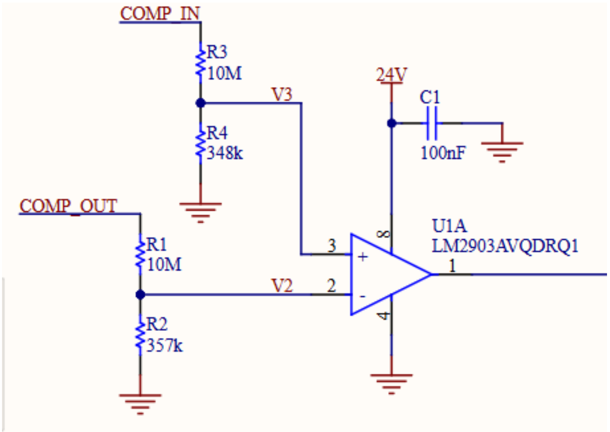

Figure 4.14: Precharge comparator.

The remaining circuitry, presented in Appendix X, consists of an optocoupler, relays, and the associated control logic implementing the precharge state machine. The system operates through three distinct states:

1. **System Off:** When either the Battery Management System (BMS) enable signal or the kill switch (KS) is inactive (in the case where the motor controller is the load), all relays remain de-energised, fully isolating the battery from the load.

2. **Precharge State:** When both the BMS enable signal and the kill switch are active, the precharge relay (K3) is energised while the main relay remains open. In this state, the load capacitors are charged through the precharge resistor, limiting the inrush current.

3. **Normal Operation:** The comparator continuously monitors the voltage difference between the battery side and the load side. Once the load-side voltage reaches approximately 98% of the battery voltage, the comparator switches state, de-energising the precharge relay (K3) and closing the main relay. The precharge resistor is thereby bypassed, and the system operates under full load voltage conditions.

This sequence ensures a controlled energisation of the capacitive load, significantly reducing inrush current and improving the reliability and lifetime of the switching components.

The precharge duration is primarily determined by the value of the precharge resistor. A higher resistance results in a longer precharge time but lower inrush current, while a lower resistance reduces precharge time at the expense of increased power dissipation. In this design, a $1\ k\Omega$ resistor was selected as a compromise between precharge time and power losses, resulting in a precharge duration of approximately 2 seconds, with an average power dissipation of 0.317 W and a peak power of 2.540 W.

4.6 PCB design

4.6.1 MPPT V1

High level explanation of the components and interconections

software used, falar sobre o design de 4 layers, pontos de debug, coisas adicionais não faladas anteriormente (pq de cada conector, porta fusiveis, serial to UART, LEDs de debug), posicionamento, segurança adicional de temperatura, dcdcs de power, can, I2C

Image of V1 and V1.1.

4.6.2 MPPT V1.1

Improvments

4.6.3 Pre charge

The Pre charge PCB was designed with the intention of being reusable across other PCB systems. For this reason, particular attention was given to achieving a compact and versatile layout. The design uses a two-layer stack-up, as the circuit is relatively simple, which reduces manufacturing complexity and overall cost while still meeting the electrical requirements. The PCB is mechanically mounted directly onto the main relay contactors, which simplifies assembly and reduces wiring complexity, Figure 4.15.

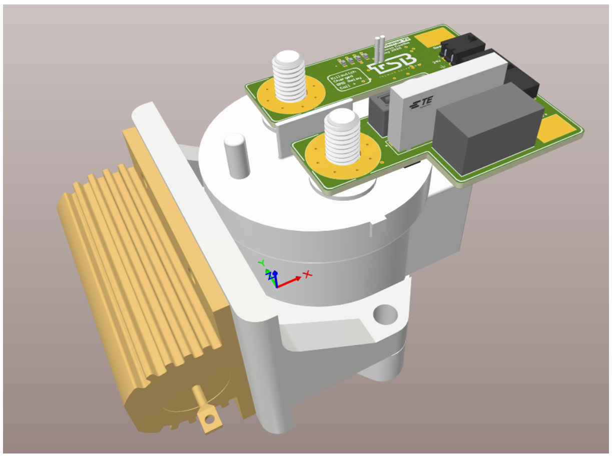

Figure 4.15: Precharge 3D design.

In addition, the pre charge resistor was intentionally placed outside the PCB, mounted to a costume 3D holder. This design choice facilitates easy replacement or adjustment of the resistor value while also contributing to a more compact board footprint. The resistor is electrically connected to the PCB by soldering wires directly to dedicated copper pads, ensuring a simple and reliable interface between the external component and the board.

5 Laboratory results

intro

5.1 Voltage and Current sensor

Both current and voltage sensors were tested in the laboratory to validate the performance and accuracy of the measurements.

The voltage sensor test was performed by applying a known voltage to the input and measuring the output of the voltage divider and the MCU readings.

The test was performed for the entire range of voltages needed for this application, which is from 0 V to 51 V. The results show that the voltage sensor is accurate and has a resolution of 15.34 mV, which is close to the theoretical value of 15.38 mV and a average error of 40 mV. However, this could be improved even further by using resistors with less tolerance and lower temperature coefficient or by increasing the ADC resolution. Another option that does not increase the cost of the system is to do a linearization of the results. In Figure 5.1 the error of the voltage readings is shown, and it can be seen that the error is linear and can be corrected by applying a linearization in the software. None of this is necessary for this application, since the resolution is already good enough for the Maximum Power Point Tracker (MPPT) operation.

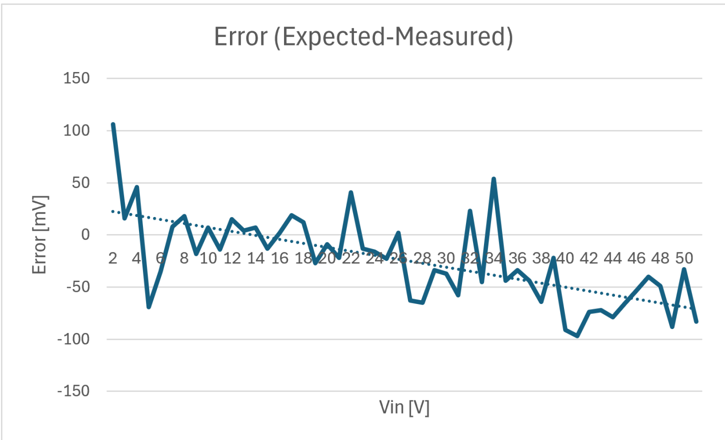

Figure 5.1: Voltage sensor performance.

The same test was performed for the current sensor, the measured resolution was 3.72 mA, which is close to the theoretical value of 3.66 mA. The average error was 52 mA, which is very good for this application. The error of the current sensor is shown in Figure 5.2, and it can be seen that the error is linear, but it is still very small and does not affect the MPPT operation.

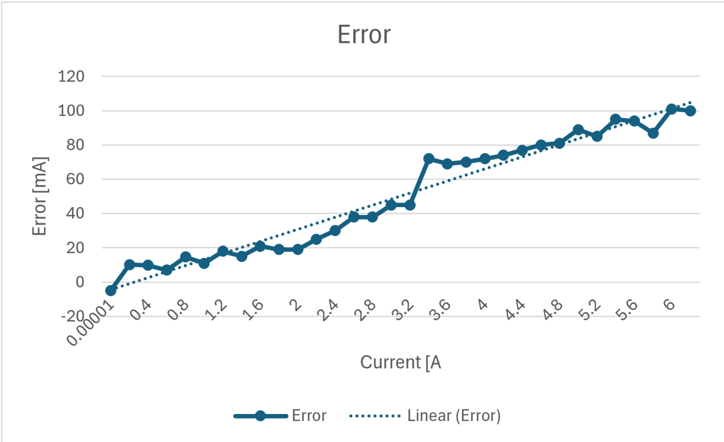

Figure 5.2: Current sensor performance.

In both cases the linearization whould highly increase the accuracy of the measurements and it is highly recommended. However, it takes a lot of time to do the measurements and calculate the linearization for each Printed Circuit Board (PCB) which is not realistic for a student prototype project like Técnico Solar Boat (TSB). So the linearization was not implemented in this project, but it is highly recommended for future work.

Tenho imagens do setup, tabelas de excel com os resultados e mais graficos que posso por aqui ou em anexos

5.2 Boost

Describe the test and want i want to achieve

5.2.1 Setup

Describe the setup used

Fonte com resistencia como load. O circuito consegue ligar divido ao body diode. Uso multimetros diitais para medições precisas, termoetros para medição de temperatura.

Dizer que há voltage drop nos cabos então a medição tem de ser feita nas entradas do MPPT

Com app facilmente se analise os resultados

Foto do setup

5.2.2 Results

5.2.2.A Switching noise

During the initial experimental tests, a significant amount of electrical noise was observed in all measured signals. This noise caused the measured current to exceed the configured Over Current (OC) protection thresholds, resulting in repeated activation of the protection mechanism. Initially, it was assumed that the problem originated from the measurement system. Consequently, several modifications were implemented in the firmware, including reducing the measurement update frequency to minimize ADC crosstalk and increasing the ADC sampling time to improve measurement accuracy. Furthermore, the fault-handling logic was modified because, once a OC fault occurred, the converter entered an oscillatory state in which it repeatedly enabled and disabled itself. Although these software modifications improved the stability and accuracy of the measurements, they did not eliminate the excessive noise.

During the OC events, a large oscillation in the input current was also observed, varying approximately between 4 A and 12 A. One possible explanation considered was that the laboratory power supply was operating close to its current limit, causing it to become unstable. Replacing the power supply was therefore considered as a potential solution. However, since the observed noise was considerably higher than expected, it was believed that the root cause was located elsewhere.

A detailed analysis of the reference evaluation board revealed a significant difference in the output decoupling network. While the initial prototype employed a single 1 μF ceramic capacitor, which completely eliminated the switching noise in simulation, the reference evaluation board used ten 1 μF ceramic capacitors connected in parallel.

To evaluate the impact of the output capacitance, the converter was characterized without any changes. Figure 5.3 shows the initial noise.

In all oscilloscope captures, Channel 1 (yellow) corresponds to the output current (I_{out}), Channel 2 (blue) to the input voltage (V_{in}), Channel 3 (orange) to the input current (I_{in}), and Channel 4 (green) to the output voltage (V_{out}). Although the screenshots are not of optimal quality, they clearly illustrate the reduction in switching noise obtained after increasing the output decoupling capacitance.

To increase the local output decoupling, nine additional 1 μF ceramic capacitors were soldered in parallel with the existing capacitor, matching the total capacitance used in the evaluation board. After this modification, a significant reduction in the measured noise was observed, as shown in Figure 5.4.

These results confirm that the insufficient output decoupling capacitance was one of the primary contributors to the observed electrical noise. Increasing the number of ceramic capacitors substantially improved the converter performance and reduced measurement fluctuations.

Although the results were already satisfactory, the reference evaluation board also includes seven additional 220 nF ceramic capacitors distributed across the output stage. Initially I also had one smaller capacitor (100 nF) and since i did not had any 220 nF with 100 V rating, i added six 100 nF capacitors. The result should be similar. These capacitors are expected to further reduce the high frequency switching noise. The results are presented in Figure 5.5.

The approximate peak noise amplitudes measured for each hardware configuration are summarized in Table 5.1. Adding nine 1 μF ceramic capacitors significantly reduced both the current and voltage noise. Subsequently, the six additional 100 nF ceramic capacitors further reduced the current noise, although the input voltage increased but it may be due to imprecise measurements from the oscilloscope. I tried to do the same measurements with a multimeter, however the precision was similar.

Table 5.1: Approximate noise amplitudes measured for the three hardware configurations.

Quantity	Original	+9 $\times$ 1 μF	+9 $\times$ 1 μF +6 $\times$ 100 nF
I_{in}	1.8 A	640 mA	292 mA
I_{out}	0.8 A	300 mA	220 mA
V_{in}	2.0 V	350 mV	380 mV
V_{out}	8.0 V	2.3 V	2.2 V

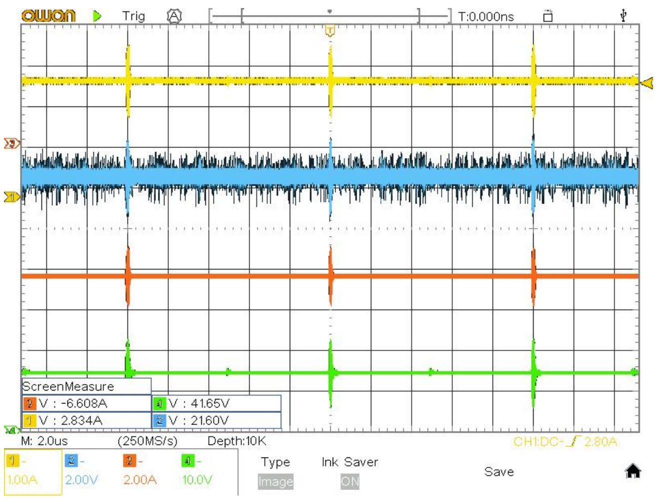

Figure 5.3: Measured waveforms before adding the additional ceramic capacitors.

5.2.2.B Temperature rise

mostra o aumento da temperatura para cada caso.

mostrar a foto termica, ter em conta que a imagem e o prefil de temperatura estão ligeiramente deslocados.

concluir que não é preciso heatsink e que os componetes que aquecem mais é a bobine. Nos outros testes medimos alguma aumento de temperatura também nos condesadores, porem a temperatura irradia e por sua vez aquece os condensadores. O que leva a algumas medições erradas. Eu tirei a

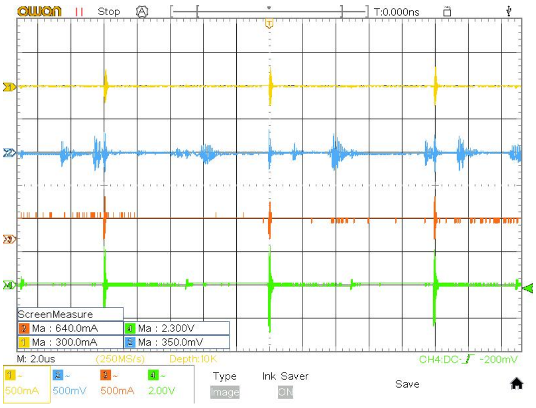

Figure 5.4: Measured waveforms after adding nine additional 1 μF ceramic capacitors.

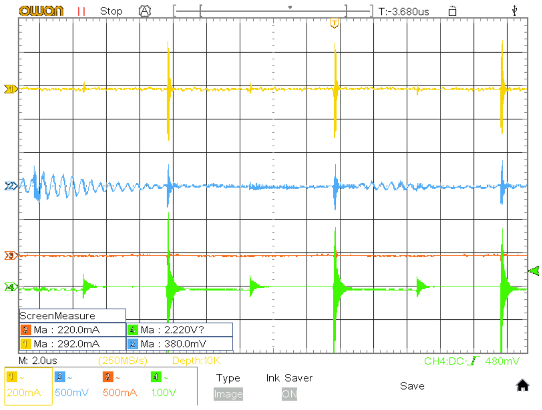

Figure 5.5: Measured waveforms after adding nine additional 1 μF ceramic capacitors.

temperatura mais alta dos condensadores mas era notavel a diferença entre os condesadores perto da bobine e os condesadores longe.

Inicialmente os condesadores aqueciam mais devido aos picos de corrente gerados pelo ruido.

É interesante notar tambem um aumento de temperatura em alguns traces. Nomeadamente a linha que contorna a PCB é onde está localizado o trace que aliemnta os dcdcs para 5V e 3.3V que alimenta toda a eletronica auxiliar da placa. Isto significa que deveria ser um tace mais largo.

O mesmo aconte na zona dos sensors de current.

Já np lado de baixo da placa, apenas observamos um aumento de temperatura no transistor (a imagem está shiftada para baixo/cima), porem muito menor do que o aumento no condesador.

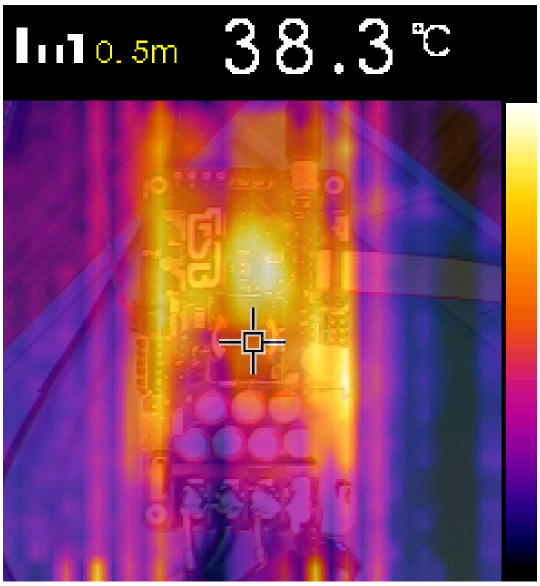

Figure 5.6

Figure 5.7

5.2.2.C Efficiency and voltage and current ripples.

Potencia perdida com o circuito desligado, + boost, e todos os casos.

cerca de 0.43w só para ter tudo ligado á volta do boost. Se ligarmos o PWM sem load esta potencia aumenta para 1.01w. O sensor de corrente não mede esta corrente e por isso mede sempre 0 A no caso de OC. (neste caso a corrente é tão baixa que praticamente não há perdas nos cabos)

Muitas perdas nos cabos. Resitencia dos cabos de entrada = 0.09 Ohm e dos cabos de saida 0.2 ohm

Efficiency, voltage and current ripple

Losses analysis

temperature analysis

5.3 MPPT

5.3.1 Setup

Describe the setup

Utilização da fonte da delta o que faz com que os resultados não dependam de fatores externos. É necessario configura la com painel solar via lan e atravez de um script que envia mensagens TCP altera a irradiancia do "painel"

a bateria pode ou não estar com problemas

5.3.2 Tracking speed

Give steps of irradiation and measure the tracking speed

5.3.3 Battery charging

Do a full charge, validate CC-CV, battery temperature, MPPT temperature and robustness

6 Conclusion

last thing to write

6.1 Future Work

last thing to write

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